

How cancer-associated fibroblasts promote T-cell exclusion in human lung tumors: a physical perspective

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eLife Assessment

This is a **valuable** report of a spatially-extended model to study the complex interactions between immune cells, fibroblasts, and cancer cells, providing insights into how fibroblast activation can influence tumor progression. The model opens up new possibilities for studying fibroblast-driven effects in diverse settings, which is crucial for understanding potential tumor microenvironment manipulations that could enhance immunotherapy efficacy. While the results presented are **convincing** and follow logically from the model's assumptions, some of these assumptions, as acknowledged by the authors, may oversimplify certain aspects in light of complex experimental findings, system geometry, and general principles of active matter research. Nonetheless, the authors provide justification for their work as a meaningful step towards more comprehensive modeling approaches.

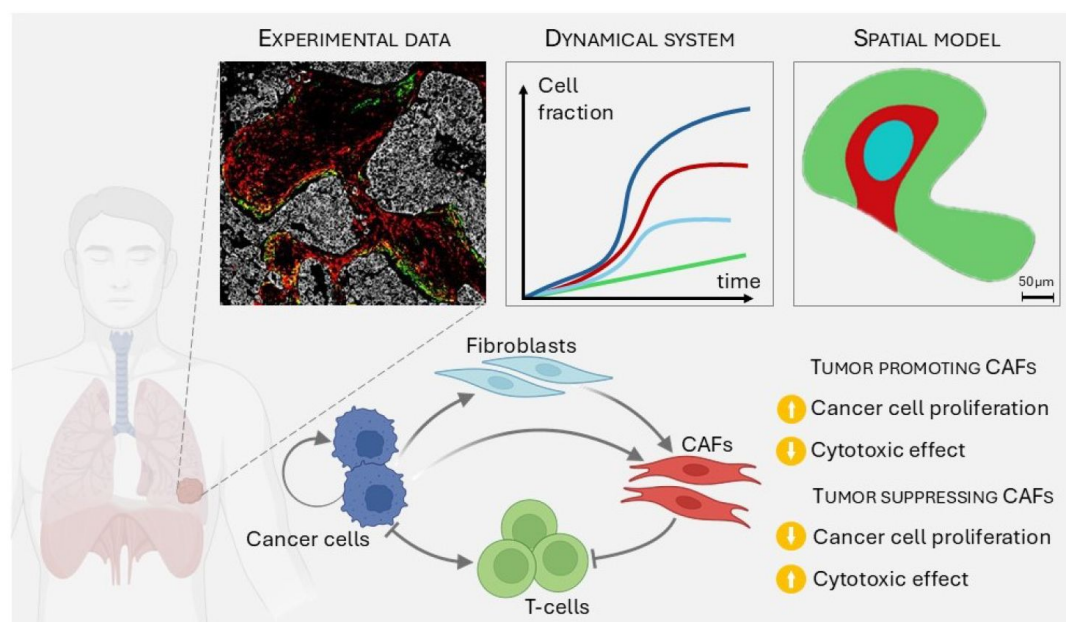
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Abstract

The tumor stroma is a tissue composed primarily of extracellular matrix, fibroblasts, immune cells, and vasculature. Its structure and functions, such as nutrient support and waste removal, are altered during malignancy. Tumor cells transform fibroblasts into cancer-associated fibroblasts, which have important immunosuppressive activity on which growth, invasion, and metastasis depend. These activated fibroblasts prevent immune cell infiltration into the tumor nest, thereby promoting cancer progression and inhibiting T-cell-based immunotherapy. To understand these complex interactions, we measure the density of different cell types in the stroma using immunohistochemistry techniques on tumor samples from lung cancer patients. We incorporate these data, and also known information on cell proliferation rates and relevant biochemical interactions, into a minimal dynamical system

with few parameters. A spatio-temporal approach to the inhomogeneous environment explains the cell distribution and fate of lung carcinomas. The model reproduces that cancer-associated fibroblasts act as a barrier to tumor growth, but also reduce the efficiency of the immune response. The final outcome depends on the parameter values for each patient and leads to either tumor invasion, persistence, or eradication as a result of the interplay between cancer cell growth, T-cell cytotoxic activity, and fibroblast attraction, activation, and spatial dynamics. Our conclusion is that a wide spectrum of scenarios exists as a result of the competition between the characteristic times of cancer cell growth and the activity rates of the other species. Nevertheless, distinct trajectories and patterns allow quantitative predictions that may help in the selection of new therapies and personalized protocols. We conclude with different options for further modeling.

Graphical abstract



Introduction

Cancer results from the malignant transformation of cells due to genetic changes or damage that causes cells to grow and spread in an abnormal and uncontrolled way. Mutated cells can form solid tumor tissue at specific sites and spread to distant regions of the body through metastasis. The growth, development, and response to drugs and therapies of tumor masses are highly dependent on biophysical environmental conditions. These involve a not fully understood crosstalk between cancer cells and the surrounding stroma, which consists of cancer-associated fibroblasts, vessels, and immune cells embedded in the extracellular matrix. Indeed, the tumor-stroma ratio turns out to be an independent prognostic factor, with a large proportion of stroma leading to a worse prognosis [1]. In this paper, we focus on modeling tumorstroma interactions in lung cancer which is one of the most common and deadly cancers, with more than 2.2 million new cases diagnosed and 1.8 million deaths worldwide in 2020. Non-small cell lung carcinoma (NSCLC) accounts for the vast majority of lung cancers (85%) [2]. The majority of cells that make up the stroma are fibroblasts and macrophages. Both types are affected and reprogrammed by cancer cells and have been shown to act as tumor promoters and tumor suppressors, depending on the stage of tumor progression and of its mutational status [3, 4]. As cancer progresses,

tumor cells often transform surrounding healthy fibroblasts into cancer-associated fibroblasts (CAFs) which, similarly to a wound healing context, produce higher levels of extracellular matrix (ECM) as well as growth factors and cytokines that affect the recruitment of immune or vascular cells and the growth of cancer cells [5].

Experimental observations and evidence of the mutual interactions between cancer cells, the immune response and cancer-associated fibroblasts suggest that a multi-physical model of the tumor microenvironment (TME), including molecular and spatial dynamics, could be highly beneficial in understanding and predicting the complex evolution of tumor growth and progression. Thus, such analysis may represent a critical step in improving T-cell-based therapies. Quantitative models already exist for biochemical interactions between fibroblasts and cancer cells [6, 7, 8], or for the T-cell recruitment in immunotherapy [9], but models for multispecies are less common. A recent paper by Mukherjee *et al.* analyzed the spatial dynamics of infiltrating splenocytes in an aggregate of cancerous melanocytes using experimental data obtained *in vitro* [10]. Their conclusion is that a strong persistent random walk and contact energies are important for the ability of T-cells to infiltrate the tumor. However, to the best of our knowledge, there are no approaches in the current literature to model the dynamics behind human lung tumor proliferation due to the role of CAFs in excluding T-cells from the nest. Here, we propose such a model for NSCLC tumors to describe the dynamic interplay between cancer cells, T-cells and fibroblasts in their non-activated and activated states, incorporating the most relevant interactions within the human tumor microenvironment.

To this end, we briefly review the literature on the interactions between different tumor components, including quantitative information on their composition. In particular, we base our model on data extracted from immuno-histochemistry (IHC) staining of the tumor microenvironment performed on a large cohort of human NSCLC tumor samples. This allows us to calibrate our model so that the densities and localization of the various species that the model yields are consistent with the actual data [11] (Section I). We then build the theoretical description in two steps. In a first step, we ignore any spatial organization and model the proliferation and activation phenomena alone, while introducing a pressure-like term to avoid excessive proliferation. This leads to a nonlinear dynamical system for the cell concentrations (Section II). Careful analysis of this dynamical system yields a stable steady state in which T-cells become inefficient against tumor survival for a given set of parameters. More precisely, we demonstrate that two mechanisms control tumor growth: the competition with other species for resources and space, and the efficiency of the T-cell response. In a second step, we enrich the model by considering the spatio-temporal evolution of tumor growth through a continuum approach for cell mixtures based on Onsager's variational principle. Indeed, it has been shown that the spatial distribution of T-cells inside the tumor is an indicator of the patient survival [12]. Finite element-based simulations in two-dimensional space demonstrate the ambiguous activity of cancer-associated fibroblasts (CAFs) in regulating cancer cell proliferation and invasion. Finally, we discuss our results obtained with this minimal model and suggest directions for future works, such as considering more precisely the geometry of the tumor micro-environment, and introducing anisotropic friction that would depend on the nematic orientation of the fibers (Section III). In the Supporting Information, further details about the results of numerical simulations are provided to show how the proposed dynamical system is able to capture different evolutionary and tumor fates depending on the role played by the species, their initial conditions, as well as the shapes and distributions of the regions occupied by cells. Appendix A is joined to this article with details for the calculation of the dynamical system section, and Appendix B for the anisotropic friction.

I. Lung Cancer Microenvironment

Modeling the tumor environment requires a deep understanding of the interactions between the various components, such as cells of different types, soft materials and fibers as well as fluids and diffusing molecules. These interactions are summarized in [section I.1](#). Next, in [section I.2](#), we present data on the density of the different cell types in the TME. Indeed, our continuous approach incorporates *in vivo* biological data that vary between patients and tumor sites, and also evolve over time within the tumor. Therefore, after a review of the available values found in the literature, we performed direct measurements from samples stained by multiplex IHC to study the composition of fibroblasts and distribution of T-cells in human

I.1 Interactions Between TME Components

Almost all tissues contain a population of fibroblasts that provide the tissue architecture, and serve as sentinels for tissue dysfunctions. When a solid tumor grows, quiescent or progenitor lung fibroblasts activate an initial wound-healing response with increased production of ECM and growth factor (cytokine) and upregulation of activation markers such as FAP [\[13\]](#). Due to mechanotransduction and/or biochemical signals from tumor or immune cells, CAFs often increase their level of contractility, which affects the maintenance of the tumor stroma [\[14\]](#). CAFs are characterized by their increased mobility, proliferation, and ECM remodeling and, unlike wound-associated fibroblasts, they seem to undergo poorly reversible activation in absence of an appropriate therapy [\[15\]](#), [\[16\]](#), [\[17\]](#), [\[18\]](#). The diversity of this population, resulting from phenotypic modifications, explains their diverse functions and localization in the stroma, close or distant from the tumor nest [\[19\]](#), [\[11\]](#).

Activated fibroblasts produce fibers that act as a mechanical barrier around the tumor, impeding the movement of immune cells and limiting the interaction between cytotoxic T-cells and cancer cells [\[20\]](#), [\[21\]](#). However, by creating such a barrier around tumors, CAFs may also prevent the spread of cancer, as mechanical stress can reduce cell spreading and promote cell apoptosis [\[22\]](#), [\[23\]](#). In addition, biochemical factors expressed by CAFs also help to modify the phenotype of T-cells or inactivate their cytotoxic capacity [\[24\]](#), [\[25\]](#), [\[26\]](#). Thus, CAFs have an ambiguous role as tumor promoters, inhibiting T-cell invasion into the nest, and as tumor suppressors, limiting the cancer growth and giving rise to an immune-excluded tumor (**Fig. 1B**), which is less proliferative than a free-growing tumor (**Fig. 1A**). Failure to confine the tumor induces high levels of cytotoxic T-cell infiltration, creating a hot tumor (**Fig. 1C**). Therefore, one aim of this study is to predict susceptibility of a tumor to T-cell infiltration according to the scheme given in **Fig. 1**.

In the context of non-small cell lung carcinoma (NSCLC), the recruitment of CD8⁺ T-cells seems to be modulated by a specific tumor-associated antigen present on the surface of the cancer cells [\[27\]](#). In particular, in the family of inflammatory proteins, also called cytokines, Interleukin-6 (IL6) and (IL8) seem, among others to stimulate the infiltration of CD8⁺ [\[28\]](#), [\[29\]](#), [\[30\]](#). The origins of CD8⁺ T-cells that infiltrate the tumor are diverse, as they differentiate from both circulating and tissue-resident precursors [\[31\]](#). However, infiltration alone does not imply that the immune response is efficient. In fact, an immunotherapy such as anti-PD1/PDL1 antibodies is often needed to boost the response of these T-cells. Chemical factors also play an important role in attracting T-cells via chemotaxis, such as the chemokines produced by dendritic cells or cancer cells [\[32\]](#). The absence of such chemicals leads to the formation of the so-called immune-desert tumor (**Fig. 1A**).

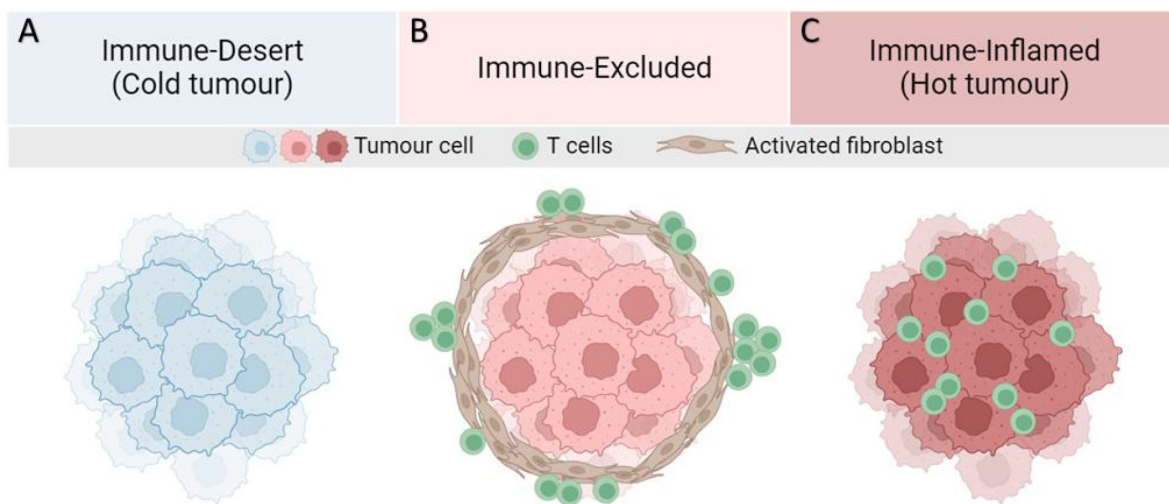


Figure 1

Major T-cell infiltration patterns observed in solid tumors.

A: Lack of tumor antigen, inadequate priming, defects in antigen presentation and/or lack of presentation, and/or lack of T-cell-attracting chemokines result in the absence of T-cells in the tumor. **B:** Presence of T-cells in invasive margins but absent in the tumor bed. Immune evasion may be due to stromal barriers, lack of chemokines, aberrant vasculature, or hypoxia. **C:** High degree of T-cell infiltration forms a hot tumor.

1.2. Lung TME Composition

For a quantitative modeling, it is critical to use biologically well-identified data on both tumor and stroma, in terms of the density and the localization of the different species. Therefore, we survey the literature to build a quantitative view for the composition of the TME. We recapitulate these values in **Table. 1** [\[10\]](#). However, human data show a great diversity depending on many factors such as tumor edge, patient age, non-cancer health status, as well as method of analysis.

The fibroblast population in the stroma, which is very heterogeneous and not easy to identify, is the best example. In particular, although fibroblasts are often considered to be a major component of the stroma, they are only found in small proportions in scRNA-seq with 10X single-cell systems, which may be rather surprising. This situation is well known and may be due to tissue digestion processes, to the fact that part of the stroma is not extracted with the tumor nodule, and to a lower efficiency of 10X single-cell systems for fibroblasts. Another caveat is that identifying the surface fraction of cells by staining can be very different from counting individual cells. In fact, cell types can vary greatly in size. For example, lung cancer cells can be between 13-18 μm [\[33\]](#), fibroblasts $\sim 16 \mu\text{m}$ [\[34\]](#), and T-cells between 5-10 μm [\[35\]](#), resulting in volumes and projected areas that can be 10 times smaller for T-cells compared to fibroblasts and cancer cells for the same densities. Moreover, in some studies, authors focus on well-characterized zones of enrichment for the different species (tumor nest, fibrotic areas), and do not consider each cell type *per se* [\[36\]](#), [\[37\]](#). Despite these remarks, cancer cells, T-cells, and macrophages (fibroblasts are underestimated as mentioned above) appear to be a significant part of the TME. The limitations of using data coming from the literature led us to perform our own measurements. In this study, we analyzed data from 13 patients with lung squamous cell carcinoma (LUSC) and 22 patients with lung adenocarcinoma (LUAD), based on a recent publication by some of the present authors [\[11\]](#). In this previous study, different fibroblast types were identified using multiplex IHC imaging. Tumor islets were stained with keratin, T-cells with CD3 and fibroblasts with αSMA , FAP and ADH1B. The coverage was evaluated for each fibroblast type and the total fibroblast coverage corresponds to the sum of these different coverages. Here, in the case of LUSC, we found that fibroblasts (composed of 6% ADH1B, 37% FAP $^+$ αSMA^- , 25% FAP $^+$ αSMA^+ , 31% FAP $^-$ αSMA^+) occupy 35% of the stroma with a tumor stroma ratio (TSR) of 1.29. In LUAD, these proportions were partly modified: fibroblasts (composed of 48% ADH1B, 18% FAP $^+$ αSMA^- , 7% FAP $^+$ αSMA^+ , 26% FAP $^-$ αSMA^+) occupy 31% of the stroma, with a TSR of 1.07. The fibroblasts most responsible for T-cell exclusion are shown to be FAP $^+$ αSMA^+ [\[11\]](#). In **Fig.2** [\[11\]](#), we compare two LUSC microenvironments. In the top one, FAP $^+$ αSMA^+ are lining the tumor (**Fig.2(a),(c),(d)** [\[11\]](#)), and T-cells are excluded from it (**Fig.2(b)** [\[11\]](#)). On the contrary, in cases where no FAP $^+$ αSMA^+ are observed, fibroblasts are homogeneously distributed in the stroma (**Fig.2(g),(e),(h)** [\[11\]](#)), and T-cells infiltrate the tumor (**Fig. 2(f)** [\[11\]](#)).

Although no data for T-cells were obtained from the last measurements, it turns out that since the surface is occupied at 80% by cancer cells and fibroblasts, and that the rest of the TME occupies less than 20%, the surface fraction of T-cells can be considered as small compared to cancer cells and fibroblasts.

II. Dynamic Modeling in the Lung Cancer TME

Our theoretical and numerical analyses consist of two steps in the spirit of Ref. [\[47\]](#) for the study of cancer stem cells. In a first analysis, we examine only the dynamics of an interactive ecological system in order to evaluate the physical parameters that quantify these interactions and how the dynamics depend on them. Spatial constraints are represented by a pressure term avoiding overcrowding. We present this approach step by step in order to set the parameters one by one, highlighting the physical importance of each choice through stability analyzes of the

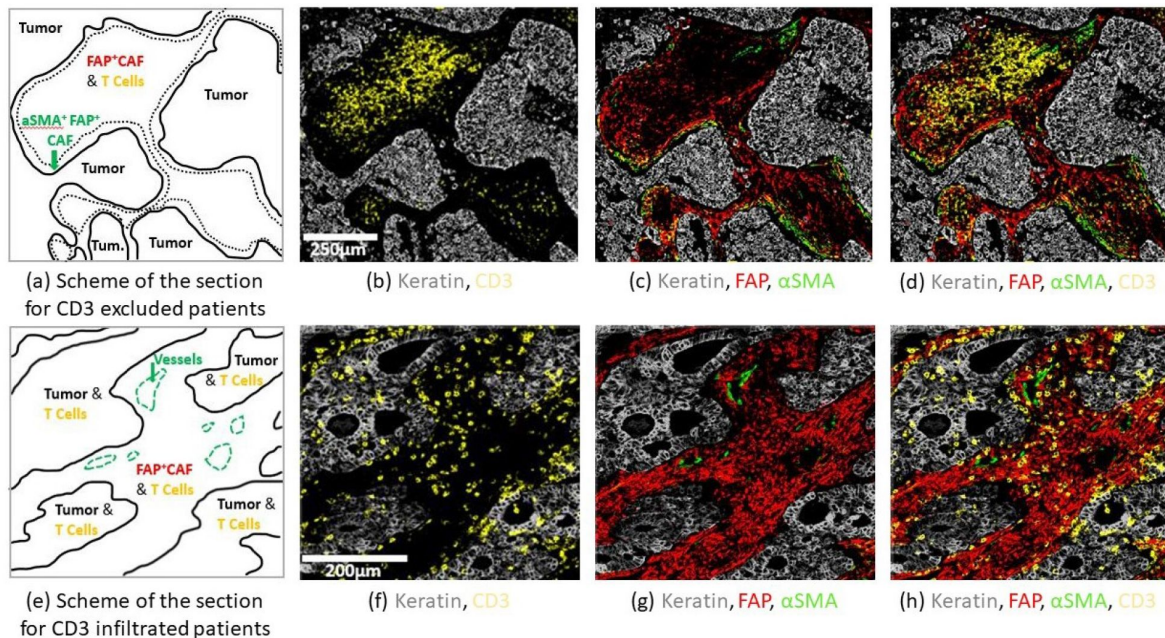


Figure 2

Structural organization in human NSCLC.

Staining was performed by multiplex IHC. FFPE NSCLC sections were stained for keratin as a marker of cancer cells (gray), CD3 (yellow), and fibroblast markers αSMA (green), FAP (red). **First row:** CD3 excluded patient. (a): scheme of the section showing CD3+ cell exclusion from tumor nests. The green arrow highlights border regions with contractile fibroblast barrier αSMA+FAP+ and low CD3+ cells. (b): CD3+ cells are localized in the center of the stroma. (c): dense αSMA staining at the tumor border are associated with a decrease of CD3+ cell abundance. (d): recap of all the markers. **2nd row:** CD3 infiltrated patient. (e): scheme of the section showing CD3+ infiltration in the tumor islets. The green arrow shows αSMA+ staining on vessels. (f): CD3+ are localized in the tumor nest. (g): FAP+ staining is localized throughout the stroma. (h): recap of all the markers.

system. In addition, this first step allows to explore efficiently the parameter space without time consuming numerical resolution, and to already produce a first classification of the global outcomes. The second step is the spatial description of the tumor growth in [Section III](#). In both approaches, the case where fibroblasts, cancer cells and T-cells are present is intended to correspond to the patient data presented in the previous section.

II.1. Dynamical System for Immune and Cancer Cells in Interaction

As we saw in the previous section, the complexity of the microenvironment makes the role of the immune system hardly predictable and highly dependent on the tumor being studied. In the case of lung tumors, the immune system is triggered as the carcinoma expands, but T-cells may be excluded from the tumor nest by activated fibroblasts. Therefore, the goal of this work is to physically and quantitatively understand the process of T-cell exclusion from the tumor mass in the simplest way possible, and to explore different possible scenarios, since the dynamical system can be seen as the spatial integration of the different processes over the domain under study.

We focus on the interaction between different cell types in the case of the NSCLC. In particular we consider a closed system including the cancer cells, T-cells, non-activated fibroblasts (NAFs), cancer-associated fibroblasts (CAFs) and healthy cells with the extracellular medium. Diffusive signaling molecules are not explicitly introduced: their production by one cell type and their effect on another cell type is modeled as a direct interaction between the two. For example, the attraction of T-cells to cancer cells by chemotaxis is introduced in the mathematical model as a source term proportional to the product of the two concentrations T and C in the T-cell equation (see below [Eq. 2](#)). We also hypothesize that the main difference between NAFs and CAFs is the fiber production of the latter, which prevents T-cells from infiltrating the tumor. Furthermore, our model does not consider transformation of NAFs into CAFs as a reversible process. All these cells have the same mass density and the sum of their mass fraction satisfies the relationship $\mathcal{S} = C + T + F_{NA} + F_A = 1 - N$, where N is a healthy non active component such as healthy cells and interstitial fluid, for example. The mass fraction of cancer cells is represented by C , T-cells by T , quiescent or non-activated fibroblasts NAFs by F_{NA} , activated fibroblasts CAFs by F_A . Note that N is not an active component and therefore does not appear in the following equations. Also, we do not consider here the recruitment of macrophages to better highlight the competing mechanisms related to the sole role of T-cells and CAFs in the tumor mass development. With this in mind, we write an evolution equation for each component of the system: $dX/dt = \Gamma_X$, where Γ_X corresponds to a source term modeling the proliferation, death, differentiation, or fluxes into/out of the system under study. The source terms for each species are described in detail below.

The dynamics of the cancer cells is driven by their proliferation, controlled by a growth rate coefficient α_C , and limited by a death rate $\tilde{\delta}_C$. It takes into account the population pressure caused by self-inhibition as well as by the inhibition of the other species. In the following, we will choose α_C^{-1} as the unit of time, and all coefficients introduced in the following will be pure constants without unit, so that $\tilde{\delta}_C = \alpha_C \delta_C$. In addition, cytotoxic T-cells eliminate cancer cells if their anti-tumor activity is not inhibited by the activated fibroblasts (although they do not remove T-cells from the mixture). This process is quantified by the cytotoxic coefficient δ_{CT} , and by the coefficient of T-cell inhibition by CAFs, δ_{TF} . With these assumptions, the dynamics of the cancer cell mass fraction can be read:

$$\frac{dC}{dt} = C - \frac{\delta_{CT}CT}{1 + \delta_{TF}F_A} - \delta_C C \mathcal{S}. \quad (1)$$

Although proliferation of cytotoxic T-cells has been observed, we do not consider explicitly proliferation in our study as we focus on their ability to infiltrate the tumor. Rather, we consider that T-cells proliferate outside the domain boundaries, so that this proliferation is included in the

boundary source contributions. Therefore, their only source is their attraction to cancer cells, which occurs at a recruitment rate of $\alpha_{TC}C$, while the crowding limits this recruitment:

$$\frac{dT}{dt} = \alpha_{TC}C - \delta_T T \mathcal{S}. \quad (2)$$

Fibroblasts are the key regulators of tumor immunity and progression. Their dynamics involve a recruitment of non-activated fibroblasts α_{NA} whose role is to maintain an adequate supply given by α_{NA}/δ_{NA} in healthy tissue, i.e. in the absence of cancer cells, and a death rate due to the pressure exerted by the cells and controlled by δ_{NA} . NAFs are attracted to the tumor nest at a rate $\alpha_{NA,C}C$ by the cancer cells that activate them to become CAFs. This process is accounted by introducing a specific transformation rate controlled by the plasticity coefficient K_A , so that the dynamic equation for NAFs reads:

$$\frac{dF_{NA}}{dt} = \alpha_{NA} + \alpha_{NA,C}C - K_A F_{NA}C - \delta_{NA} F_{NA} \mathcal{S}. \quad (3)$$

Thus, the dynamics of the CAFs is purely determined by the transformation of the NAFs and by the pressure through δ_A , which leads to:

$$\frac{dF_A}{dt} = K_A F_{NA}C - \delta_A F_A \mathcal{S}. \quad (4)$$

Therefore, the role of fibroblasts in human lung carcinoma can be investigated by studying the interplay between cancer cells and the TME. The interactions discussed above are described by 11 parameters and lead to a system of 4 coupled nonlinear differential equations concerning 4 unknowns. Within this framework, we proceed to the estimation of the parameters and the steady states of the system.

II.2 Model Parameters and Fixed Point Analysis

When all the parameters vary, the steady states or the equilibrium points of the dynamics can be quite impractical, so we start by assuming that all the coefficients at the origin of a pressure are equivalent: $\delta = \delta_C = \delta_T = \delta_A = \delta_{NA} > 0$. This reduces the number of independent parameters to 7. Fixed points are obtained by setting the time derivatives in Eqs. (1)-(4) to 0 which gives the long-term behavior when the system reaches equilibrium.

In order to evaluate the values of the parameters in Eqs. (1)-(4), we study different simplified situations that can be reproduced in experiments *in vitro* or in controlled experiments *in vivo*. We also use the conclusions obtained in Section. I.2 for the cell densities. We will start by analyzing the case of cancer cells alone and we will successively add all the other cell types with fractions C , T , F_{NA} , F_A . More details on the calculations can be found in Appendix A.

II.2.1. Cancer cells and T-cells

The dynamic evolution of cancer cells alone is limited to: $dC/dt = C - \delta C^2$. This equation leads to two equilibrium fixed points: $C = 0$ and $C = \delta^{-1}$ and only the second one $C = \delta^{-1}$ is stable. When cancer cells are isolated from other active cellular components, they are expected to invade the system or to be its major component, leading to a cancer death rate of $\delta \sim 1$.

We now examine the interaction between T-cells and cancer cells in the nest in the absence of fibroblasts (so $F_{NA} = F_A = 0$) and the relevant parameters scaled by δ are δ_{CT} and α_{TC} . Then, we study the equilibrium regime:

$$\begin{cases} \frac{dC}{dt} = 0 = C - \delta_{CT}CT - \delta C(C + T), \\ \frac{dT}{dt} = 0 = \alpha_{TC}C - \delta T(C + T). \end{cases} \quad (5)$$

There are three equilibrium solution pairs $\{C, T\}$, including the trivial one: $\{0, 0\}$. To analyze whether the solutions found are physically relevant ($0 \leq C, T \leq 1$) and dynamically stable, we estimate possible scaling for the two parameters, focusing on an effective immune response against cancer. For efficient elimination of cancer cells by T-cells, the killing rate δ_{CT} must be much larger than the natural death rate δ , so we introduce a small parameter $0 < \epsilon \ll 1$, so that $\delta_{CT} = \delta\epsilon^{-1}$. We also assume that the T-cell recruitment is slow compared to cancer cells, which means that $\alpha_{TC} = a_0\epsilon$, where a_0 being of order one. So the only stable solution is $\{C_+, T_+\} = \frac{\epsilon}{\delta} \left\{ \frac{1}{a_0 - 1}, 1 \right\}$ where $a_0 > 1$ (see Appendix A.1). Thus, even if the inflammation level is low, resulting in a small number of T-cells, the immune action on the cancer cells remains efficient.

II.2.2. Role of activated fibroblasts on T-cells

Fibroblasts play an active role in the exclusion of the T-cells from the tumor nest. We isolate a subsystem composed of cancer cells, T-cells and active fibroblasts to determine the inhibition rate δ_{TF} , responsible for the marginalization of the T-cells and subsequently for the increase of cancer cells at a fixed volume fraction of fibroblasts. We consider the fixed points corresponding to an ensemble $\{C, T, 0, F_A\}$. For simplicity, we first assume that F_A is constant, which leads to the dynamical system:

$$\begin{cases} \frac{dC}{dt} = 0 = f_0C - \Delta_{CT}CT - \delta C(C + T), \\ \frac{dT}{dt} = 0 = \alpha_{TC}C - \Delta_F T - \delta T(C + T), \end{cases}$$

where $f_0 = 1 - \delta F_A$, $\Delta_{CT} = \delta_{CT}(1 + \delta_{TF}F_A)^{-1}$ and $\Delta_F = \delta F_A$. There are four solution pairs (see Appendix A.2).

We use the scaling of both parameters already established in the previous paragraph: $\delta_{CT} = \delta\epsilon^{-1}$, $\alpha_{TC} = a_0\epsilon$. We also assume that the inhibition of the T-cells may counteract their cytotoxic effect on cancer cells, i.e. $(1 + \delta_{TF}F_A)^{-1} \sim \epsilon$, when activated fibroblasts are abundant $F_A = \delta^{-1}f_a$. This results in $\delta_{TF} = d_0\delta\epsilon^{-1}$.

The only stable equilibrium solution is $(C_+, T_+) = (f_0\delta^{-1} + \mathcal{O}(\epsilon), \mathcal{O}(\epsilon))$, where the notation $\mathcal{O}(\epsilon)$ denotes a quantity whose order of magnitude is ϵ . Thus, when fibroblasts inactivate T-cells, they promote the cancer cell proliferation.

II.2.3. Residual Fibroblasts in Healthy Tissue

In healthy tissue, most of the fibroblasts are in a quiescent state and are not activated in the absence of pathologies such as wounds, allergic reactions or cancer cells. Therefore, in such tissues, for a quiescent fibroblast population, the density of F_{NA} is the equilibrium solution of the equation $dF_{NA}/dt = 0 = \alpha_{NA} - \delta F_{NA}^2$. The parameter α_{NA} represents the net influx of fibroblasts into the tissue. Assuming the fraction of non-activated fibroblasts to be residual $\sim \epsilon^2$, we deduce $\alpha_{NA} \sim \delta\epsilon^4$.

II.2.4. Fibroblast Plasticity

Fibroblast plasticity is the phenotypic change responsible for T-cell inhibition and for more active fiber production. Cancer cells drive this transformation of the current population F_{NA} into F_A . This process is quantified by the constant K_A . We first estimate that the fibroblast population is comparable to the cancer cell population when they are alone, which leads to: $F_{NA} = f_n \delta^{-1}$ where $f_n \sim 1$ is a constant.

Replacing $\delta_{CT} = \delta \epsilon^{-1}$, $\alpha_{TC} = a_0 \epsilon$, $\delta_{TF} = d_0 \delta \epsilon^{-1}$, leads to:

$$\begin{cases} \frac{dC}{dt} = 0 = C - \frac{\delta \epsilon^{-1} CT}{1 + d_0 \delta \epsilon^{-1} F_A} - \delta C \mathcal{S}, \\ \frac{dT}{dt} = 0 = a_0 \epsilon C - \delta T \mathcal{S}, \\ \frac{dF_A}{dt} = 0 = K_A f_n C / \delta - \delta F_A \mathcal{S}. \end{cases} \quad (6)$$

Because of the effect of the fibroblasts on them, the T-cells have little effect on the cancer cells, so cancer cell proliferation is little affected. At equilibrium, this leads to $C \sim \delta^{-1}$, $T \sim \epsilon \delta^{-1}$. The equation for T gives $F_A \sim a_0 \delta^{-1} / d_0$. This means that the activated fibroblasts must be relatively abundant to fully inhibit the activity of the T-cells. It follows that the plasticity parameter is of low order in ϵ , since the non-activated fibroblasts are maintained at a high density: $K_A \sim \delta$ (see Appendix A.3).

II.2.5. Tumor Fibroblast Attraction

In this last step, we want to determine the parameter that controls the attraction of the fibroblasts to the tumor. To do this, we write $K_A = k \delta$ in the system of equations Eqs. 1 [↗](#) → 4, which we rewrite according to the previous findings:

$$\begin{cases} \frac{dC}{dt} = 0 = C - \frac{\delta \epsilon^{-1} CT}{1 + d_0 \delta \epsilon^{-1} F_A} - \delta C \mathcal{S}, \\ \frac{dT}{dt} = 0 = a_0 \epsilon C - \delta T \mathcal{S}, \\ \frac{dF_{NA}}{dt} = 0 = a_1 \delta \epsilon^4 + \alpha_{NA,C} C - k \delta F_{NA} C - \delta F_{NA} \mathcal{S}, \\ \frac{dF_A}{dt} = 0 = k \delta F_{NA} C - \delta F_A \mathcal{S}. \end{cases} \quad (7)$$

Then, we look for solutions of the type: $\{C, T, F_{NA}, F_A\} = \delta^{-1} \{c_0, T_0 \epsilon, f_{NA}, f_A\}$. The order of magnitude of the attraction parameter is then $\alpha_{NA,C} \sim 1$ (see Appendix A.4).

In the next section, we summarize all scaling laws possible, according to the obtained results and present the different scenarios related to the state of the tumor, shown in [Fig.1](#) [↗](#).

II.2.6. Numerical study of cell population dynamics

The dynamics of each cellular component of the mixture can be systematically studied according to the full set of parameters summarized in [Table 2](#) [↗](#), with the corresponding orders of magnitude. Some of them can be considered as fixed in the system, i.e. they do not vary significantly between the different tumors. This category includes δ^{-1} , the inverse of the free tumor cell density, and α_{NA} , the NAF attraction parameter to healthy tissue. The other parameters can be studied as control parameters.

Table 1

Composition of the tumor microenvironment.

This table presents the literature about the fraction of different species in the lung TME. The values found in the present studies are written in the two last rows. Different methods have been used, in different subtypes of lung cancer. We introduce the following abbreviations. (C-c): Cancer cells. (M Φ): Macrophages. (T-c): T-cells. (Fb): Fibroblasts. (scRNA-seq): single-cell RNA seq,(NSCLC): Non Small Cell Lung Cancer, (LUAD): Adenocarcinoma, (SSN): Sub-Solid Nodule. (S): stroma region. (T): tumor nest region. (ST): Stroma+tumor nest region. (TSR): tumor stroma ratio. Data for Ref. [38, 39, 40, 41, 42] were extracted from [43]

Method	Sample	C-c (%)	T-c (%)	Fb (%)	M Φ (%)	TSR
Staining [36]	LUAD	>25		>20		0.3
Staining [37]	LUAD	67		>6.5		2.6
scRNA-seq [44]	NSCLC		55 (S)	4 (S)	15 (S)	
scRNA-seq [38]	SCLC	76				
scRNA-seq [39]	LUAD	33	19.5	~ 6	20.4	
scRNA-seq [40]	LUAD	<12	42.6	~ 1	12.2	
scRNA-seq [41]	NSCLC	<22.7	16	8	23	
scRNA-seq [42]	mixed	19.9	28.1	~ 2	27.5	
scRNA-seq [45]	LUAD	16.4	30	~2	18.4	
scRNA-seq [46]	LUAD	12	7.5			
Staining [5]	NSCLC			75 (S), 0 (T)		
Staining	LUAD	47 $\pm$ 13		31 $\pm$ 17 (S), 0 (T)		1.07 $\pm$ 0.68
Staining	LUSC	54 $\pm$ 9		35 $\pm$ 12 (S), 0 (T)		1.29 $\pm$ 0.51

Table 2

Scaling variation and estimation of the coefficients according to different scenarios.

The scenarios are described in Section II.2 and shown in Fig. 3. The coefficients above are those introduced in Eqs. (1)-(4). The left column summarizes the different roles that T-cells can play in a cell mixture and the values of the coefficients of the mixture are listed in the following horizontal line of the table. The scaling of δ is always 1, and $\alpha_{NA,C} \sim \delta\epsilon^4$.

	δ_{CT}	α_{TC}	δ_{TF}	$\alpha_{NA,C}$	K_A
Control of:	Killing C-c by T-c	Attraction of T-c by C-c	Inhibition of T-c by F-c	Attraction of F-c to C-c	Activation of F-c
Efficient T-cells (T-c) but no attraction by (C-c)	$\delta\epsilon^{-1}$ ~ 10	ϵ^2 ~ 0.01	No role	No role	No role
Efficient T-cells but inhibited by (F-c)	$\delta\epsilon^{-1}$ ~ 10	ϵ ~ 0.1	$\delta\epsilon^{-1}$ ~ 10	1 ~ 1	δ ~ 1
Inefficient T-cells no need of fibroblasts	δ ≤ 1	ϵ ~ 0.1	No role	No role	No role
Efficient T-cells not inhibited by fibroblasts	$\delta\epsilon^{-1}$ ~ 10	ϵ ~ 0.1	δ ~ 1	1 ~ 1	δ ~ 1
Efficient T-cells and fibroblasts not attracted	$\delta\epsilon^{-1}$ ~ 10	ϵ ~ 0.1	$\delta\epsilon^{-1}$ ~ 10	ϵ ~ 0.1	δ ~ 1
Efficient T-cells fibroblasts not activated	$\delta\epsilon^{-1}$ ~ 10	ϵ ~ 0.1	$\delta\epsilon^{-1}$ ~ 10	ϵ ~ 1	$\delta\epsilon$ ~ 0.1

Parameter	value	Parameter	value
D_0	$10^6 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{T}_0^{-2}$	δ	1.18 T_0^{-1}
D_C	$10^5 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{T}_0^{-2}$	τ_T, τ_{NA}	$2.5 \times 10^{-3} \mu\text{m} \cdot \text{T}_0^{-1}$
D_T, D_{NAF}, D_{CAF}	$2.5 \times 10^5 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{T}_0^{-2}$	δ_{CT}	5 T_0^{-1}
λ_{CC}	$1.25 \times 10^6 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{T}_0^{-2}$	ξ_0	$3 \times 10^2 \text{ kg} \cdot \mu\text{m}^{-3} \cdot \text{T}_0^{-1}$
λ_{CT}	$7.5 \times 10^5 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{T}_0^{-2}$	ξ_1	$3 \times 10^5 \text{ kg} \cdot \mu\text{m}^{-3} \cdot \text{T}_0^{-1}$
λ_{CF}	$0 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{T}_0^{-2}$	K	$10 \text{ T}_0^{-1} \cdot \mu\text{m}^2$
κ	$3.6 \times 10^7 \text{ kg} \cdot \mu\text{m} \cdot \text{T}_0^{-2}$	λ_{cC}	$30 \mu\text{m}$

Table 3

Values of the parameters in the spatial model

The simulated time-dependent densities of each cell type are displayed in [Fig. 3](#) for different sets of parameters in the system of equations in [Eqs. \(1\)-\(4\)](#). At time $t = 0$, we assume small mass fractions of cancer cells, T-cells, activated and quiescent fibroblasts. Over time, the particular choice of a quadratic model allows the dynamics to reach a plateau for each cell type, confirming the stability of the fixed points found in [Section II.2](#).

In the case of an immune-desert tumor, i.e. when T-cells are not attracted to the tumor nest or are unable to penetrate it ([Fig. 3\(a\)-3\(d\)](#)), cancer cell growth is not limited by the immune response. However, this growth saturates when it reaches a steady state due to the competition for space and resources between the different species and controlled by δ and $\alpha_{N_{A,C}} + \alpha_{TC}$.

T-cells efficiency can be quantified through the parameter $\alpha_{TC} \delta_{CT} / (\alpha_{N_{A,C}} K \delta_{TF})$ ([Fig. 3\(i\)](#)). If the T-cells are efficient, the tumor is said to be immune-inflamed and several scenarios can be discussed. Their response is triggered by the proliferation of cancer cells. Thus, in the absence of CAF inhibition, T-cells inhibit cancer growth ([Fig. 3\(f\)-3\(h\)](#)). In contrast, in the immune-excluded tumor, CAFs impede the T-cell response and thus indirectly promote cancer cell growth, as observed in [Fig. 3\(e\)](#).

In conclusion, our dynamical system, limited to 4 cell types, recapitulates the different possible scenarios that could evolve according to the interaction between cancer cell, T-cells and fibroblasts in lung adenocarcinoma. With respect to cancer cells, three main outcomes can be identified: the tumor can invade the tissue ([Fig. 3\(a\)](#)), the cancer cell population can be limited without disappearing ([Fig. 3\(b\)-3\(e\)](#)), or it can be eradicated ([Fig. 3\(f\)-3\(h\)](#)). Each outcome can have multiple origins, that are all related to either competition for space and resources or T-cell efficiency, see [Fig. 3\(i\)](#). Each origin has a particular signature with respect to the different species densities, so that the TME properties can be inferred from the measured densities. However, the dynamical system does not provide information about the morphology of the tumor, and its local composition. Indeed, this analysis poorly represents the structure of the tumor, which is divided into a nest, a cancer-associated stroma, and a healthy stroma.

For these reasons, in what follows we extend our model by incorporating the effects of cell species diffusion in order to obtain a more faithful spatio-dynamical description of the system, thus re-analysing the cases above mentioned in this more general framework.

III. Spatio-Temporal Behavior of Tumor Growth

Spatial study is an important tool in cancer diagnosis since the tumor shape reveals the aggressiveness of cancer cells and the role of their microenvironment [[48](#), [49](#), [50](#), [51](#), [52](#), [53](#), [54](#), [55](#)]. Similarly, the localization of immune cells in the tumor environment [[10](#)], as well as proliferating and pre-metastatic cells often found in niches, are active areas of research in oncology and a valuable support for clinical prognosis [[56](#), [57](#), [58](#)].

In the previous section, we described our cell mixing through the global time evolution given by a dynamical system. Here, we aim to complete the modeling by considering the spatial heterogeneity of the interplay between fibroblasts, T-cells and cancer cells and its consequences for tumor cell localization and proliferation. We first present the mixture model [[52](#), [59](#)], which is able to incorporate the spatial distribution and evolution of active cells leading to a set of partial differential equations that we solve with the finite element (FEM) software COMSOL Multiphysics (COMSOL Multiphysics® v. 6.2. www.comsol.com. COMSOL AB, Stockholm, Sweden.). Last, we propose different mechanisms as sources of anisotropy in the tumor nest, including the introduction of a nematic tensor for the orientation of fibroblasts.

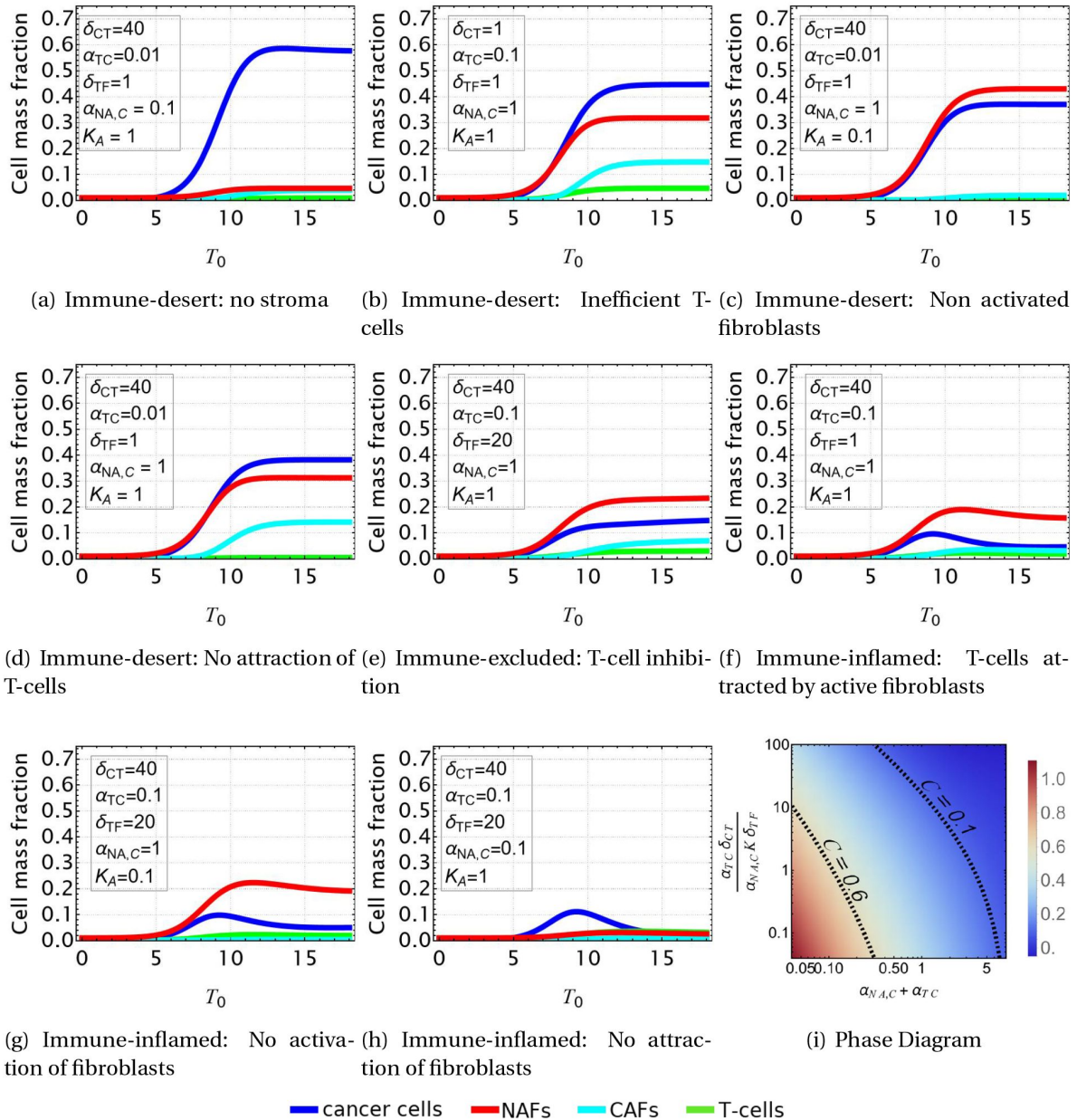


Figure 3

Evolution of the TME composition over time.

Different profiles are obtained from Eqs. (1)-(4) according to the set of parameters reported in Table. 2 and consistent with the scenarios shown in Fig. 1. In all the plots, $\delta = 1$ and $a_{NA} = 10^{-4}$. **a,b,c,d**: Immune-desert tumor. Cancer cells proliferate when the immune response is inefficient or when T-cells are not attracted to the tumor. **e**: Immune-excluded tumor. The CAF barrier inhibits T-cell infiltration, and promotes tumor growth. **f,g,h**: Immune-inflamed tumor. T-cell infiltration limits cancer cell growth. **i**: Density plot of the equilibrium cancer cell fraction at fixed δ and a^{NA} , in function of the two parameters that control the growth: $a_{TC} \delta_{CT} / (a_{NA,C} K \delta_{TF})$ reflects the ability of T-cells to kill the tumor, and $a_{NA,C} + a_{TC}$ reflects the competition between species for space and resources.

As in the previous section, we consider 4 different cell types namely cancer cells, T-cells, activated and non-activated fibroblasts, with different attraction properties. A component that does not play an active role in the mixture is also added via an inactive N fraction. It concerns the intercellular fluid, the healthy cells and the dead cells. This last component is also a source of material, for the proliferation of cancer cells. Each component is described by a local mass fraction ϕ_i , a velocity $\mathbf{v}_i$, and a proliferation/death term Γ_i , with ϕ and Γ denoting generic quantities that are now space and time dependent. More specifically, ϕ_i corresponds to the local value of C , T , F_A , F_{NA} introduced in the previous sections and $\phi_0 = 1 - \sum_{i \neq 0} \phi_i$ represents N . Γ_i is the growth rate of each component, which is now space and time dependent (see Eqs. (1)–(4)) and may be positive only for cancer cells, since we consider that fibroblasts and T-cells are recruited from the surrounding environment. CAFs produce a significant amount of fibers, resulting in a higher friction between different species, chosen to be proportional to the local amount of CAFs. This friction, which is also a space-time dependent quantity, will impact the dynamics of the mixture.

The sample also contains diffusive signaling molecules that are at the origin of the immune activity. In the previous section, chemicals were not introduced, although they were associated with some coefficients of the dynamical system. Here, the chemicals that determine the cell behavior, are represented by a concentration c [47, 60]. Note that these chemicals have no mass and diffuse through the mixture with the diffusion coefficient D_c . For simplicity, we restrict ourselves to a single chemical of concentration c , that mediates both chemotaxis and activation of fibroblasts. The balance between its production and degradation rate writes $(\tau_c^{-1} \phi_C - \delta_c) c$, since it is produced by cancer cells and naturally degraded. With these considerations, we study the case where the tumor is well supplied with nutrients, which are therefore not explicitly mentioned, and we write a set of conservation equations for each component of the mixture:

$$\begin{cases} \partial_t \phi_i + \nabla \cdot (\phi_i \mathbf{v}_i) = \sum_j \Gamma_{ij} \phi_j, \\ \partial_t c = D_c \nabla^2 c + \tau_c^{-1} \phi_C - \delta_c c. \end{cases} \quad (8)$$

When integrated over the whole space and when the only boundary conditions for the velocity are considered, the first equation in (8) recovers the dynamical system. Besides, we consider that chemicals equilibrate much faster than the tissue dynamics, so that: $\lambda_c \nabla^2 c + a_{cc} \phi_C - c = 0$, with $\lambda_c = \sqrt{D_c / \delta_c}$ the penetration length and $a_{cc} = (\delta_c \tau_c)^{-1}$. We now present a derivation for the average local velocity $\mathbf{v}_i$ of each species by evaluating its momentum equation and the expressions for the various source terms.

III.1 Momentum and Free Energy Density Derivation

Since the dynamics is very slow and completely controlled by dissipation, the Onsager variational principle of least dissipation [61] yields the set of partial differential equations coupling the densities to the velocities in the mixture [59]. This principle, introduced by Lord Rayleigh and further developed by Onsager [62, 63, 64, 65], is widely used in soft matter [66], in the biophysical context [59, 52, 67, 68, 61] as well as other areas of physics [69]. First we define a free energy functional $\mathcal{F}$ from a free energy density F integrated over the volume V , the associated chemical potentials μ_i , and a dissipation function $\mathcal{W}$:

$$\begin{cases} \mathcal{F} = \int dV F(\{\phi_i\}, c) \quad ; \quad \mu_i = \frac{\delta \mathcal{F}}{\delta \phi_i}, \\ \mathcal{W} = \int \sum_{i \neq j} \frac{\xi_{ij} \phi_i \phi_j}{2} (\mathbf{v}_i - \mathbf{v}_j)^2 dV, \end{cases}$$

where the ξ_{ij} are the relative friction coefficients between components i and j , and $\mathbf{v}_i$ and $\mathbf{v}_j$ are the velocities of the different cell types. The Rayleighian is then defined as the sum of the dissipation function $\mathcal{W}$ and the rate of change of the free energy function $\mathcal{F}$. Within this framework, the final equations for the local velocities are obtained by minimizing the Rayleighian with respect to each velocity $\mathbf{v}_i$:

$$\mathcal{R} = \frac{d\mathcal{F}}{dt} + \mathcal{W}, \quad \frac{\delta \mathcal{R}}{\delta v_i} = 0 \Rightarrow \sum_j A_{ij} \mathbf{v}_j = -\nabla \mu_i,$$

where A_{ij} is the friction matrix and μ_i are the chemical potentials. Defining: $\tilde{\phi}_i = \phi_i \phi_0^{-1}$ friction matrix reads:

$$\begin{cases} A_{ii} = \sum_{j \neq i, 0} (\xi_{ij} + \xi_{j0} \tilde{\phi}_i) \phi_j + \xi_{i0} (1 + \tilde{\phi}_i)^2, \\ A_{ij} = \phi_j (-\xi_{ij} + \xi_{i0} (1 + \tilde{\phi}_i) + \xi_{j0} (1 + \tilde{\phi}_j) + \sum_{k \neq i, j, 0} \xi_{k0} \tilde{\phi}_k). \end{cases}$$

Importantly, in our model we assume that all frictions ξ_{ij} are equal, except for the friction with the CAFs. In fact, the mass fraction of CAFs is assumed to reflect the amount of matrix produced by the fibroblasts, resulting in a very high friction in the medium. Therefore, we write: $\xi_{ij} = \xi_0 \xi_{iCAF} = \xi_0 + \xi_1 \phi_{CAF}$.

After deriving the momentum equations, we construct the free energy density. Building a free-energy density for a biological material is justified, because, although biological materials are out of equilibrium, their behavior often resembles that dictated by thermodynamics. It is therefore useful to write a free energy in terms of state variables. Following [59] and inspired by the Cahn-Hilliard approach, we define a free energy density F as a sum of the interaction potential f and of the cost induced by the mass fraction gradients $\kappa(\nabla \phi_i)^2/2$. We also assume for f , the Flory-Huggins free energy density of mixing that depends on the local mass fractions (or equivalently the volume fractions) of each component ϕ_i [70, 71]:

$$\begin{cases} F = f + \sum_{i,j} \frac{\kappa}{2} (\nabla \phi_i)^2 & ; \quad \mu_i = \partial f / \partial \phi_i - \kappa \nabla^2 \phi_i, \\ f = \sum_i D_i \phi_i \log(\phi_i) - \sum_{i,j} \alpha_{ij} \phi_i \phi_j. \end{cases} \quad (9)$$

The terms $D_i \phi_i \log(\phi_i)$ control both the diffusion and volume exclusion, and the quadratic expansion $-\alpha_{ij} \phi_i \phi_j$ controls the attraction or repulsion between species.

In the following, we explain in more detail the effective free energy density of our system, as well as the expression of the different proliferation rates, and the values of the parameters, focusing on the attraction/repulsion and the chemotactic terms.

III.2. Effective Free Energy Density and Source Terms

We now apply the general formalism presented in the previous section to the cancer cell mixture using the same notation as Section II for C , T , F_{NA} , F_A . However, whereas in the first section these quantities represented averaged mass fractions over the whole tumor, here they are local mass fractions averaged over a small volume of the tumor but large compared to the cell size. We now detail the effective free energy, focusing first on the interaction potential f defined in Eq. 9.

The attraction between cancer cells is represented by $-\lambda_{CC}C^2$, between cancer cells and T-cells by $-\lambda_{CT}CT$, and between all types of fibroblasts and cancer cells by $-\lambda_{CF}C(F_{NA}+F_A)$, where the λ coefficients are positive. These interactions are accounted for the free energy density that reads:

$$f = D_C C \log(C) + D_T T \log(T) + D_{CAF} F_A \log(F_A) + D_{NA} F_{NA} \log(F_{NA}) + D_0 \phi_0 \log(\phi_0) - \lambda_{CC} C^2 - \lambda_{CT} CT - \lambda_{CF} C(F_{NA} + F_A). \quad (10)$$

Finally, we focus on the proliferation rates to complete the system Eq. 8. Note that we must distinguish the cells produced *in situ*, such as the cancer cells C and the activated fibroblasts F_A , from the cells attracted to the tumor nest (T and F_{NA}) by the chemical c . We keep the growth rate of cancer cells Γ_C given by Eq. 1 with $\delta_{TF} = 0$ because the inhibition of T-cells by fibroblasts is now treated by increasing the friction created by the fiber barrier. T-cells and NAFs do not proliferate and are not derived from precursors at the tumor site, but are instead generated far away from the tumor and are attracted to it by chemotaxis. Thus, at the boundaries S , their source rate is driven by an incoming flux due to the chemical c . For the NAFs, the volume term $\Gamma_{NA}|_V$ does not include the source term $\alpha_{NA,C}$ (see Eq. 3) which is provided by the boundaries, and α_N is neglected:

$$\begin{cases} \Gamma_T|_S = \frac{c(1-\mathcal{J})}{\tau_T} & \text{and} & \Gamma_T|_V = -\delta T \mathcal{N}, \\ \Gamma_{NA}|_S = \frac{c(1-\mathcal{J})}{\tau_F} & \text{and} & \Gamma_{NA}|_V = -K_A F_{NA} C - \delta F_{NA} \mathcal{S}. \end{cases}$$

On the other hand, the production of CAFs is not changed and Γ_A defined in Eq. 4 is still relevant, provided that the plasticity contribution is now proportional to the chemical concentration c and not the cancer cells fraction.

We scale the various physical parameters in Table. 3. Their values are derived from the literature and from the spatial structure of the TME, as well as the expected outcomes for the different scenarios, while the growth parameters are related to those in the first part of the article. For example, the tumor nest is a dense phase of cancer cells with mass fraction $C = 0.8$ that an interface of size $d \sim 10 \mu\text{m}$ separates from the dilute phase $C \sim 0$. This choice, together with the typical value for the energy-density of $10^6 \text{ kg} \cdot \mu\text{m}^{-1} \cdot \text{d}^{-2}$ found in literature imposes the values of $D_C, D_0, \lambda_{CC}, \kappa$ [59]. Similarly, we assume a weak infiltration of T-cells and fibroblasts into the nest, resulting in low values for the attraction parameters $\lambda_{CT}, \lambda_{CF}$ when compared to λ_{CC} . Since D_{NA}, D_{CAF}, D_T control the diffusion of the fibroblasts and T-cells, whose source is located at the boundaries, their values are determined by the density gradients between the boundaries and the tumor nest. At the same time, the times required for T-cells to kill the tumor nest is given by δ_{CT} , and the times for the arrival of the T-cells and NAFs into the domain, as well as their average mass fractions, are governed by τ_T and τ_F , respectively. The NAFs transform into CAFs in a time controlled by K . We use the letter T_0 as the time unit which corresponds to the typical division time for tumor development. For experiments *in vitro* in perfect conditions of nutrient access, the relevant time is the largest one, corresponding to the slowest process, so the typical division time of cancer cells. It is about one day and the same time is needed for the eradication of tumor cells [72, 73, 74]. But *in vivo*, it is obviously a very different time. It is more related to the time scale necessary for the tumor to grow in its natural environment and is of the order of months [75]. The plasticity of fibroblasts is controlled by the chemical penetration length. Since we assume that only the fibroblasts lining the tumor nest are activated, we choose a low penetration length of $30 \mu\text{m}$. At the same time, the ratio α between the chemical production and degradation parameters is set to $\alpha_{CC} = 3$.

In the following, we will show that, once activated by tumor cells, fibroblasts can inhibit tumor growth through a confinement effect but also limit the cytotoxic role of T-cells or simply prevent their infiltration into the tumor.

III.3. Ambiguous Role of Fibroblasts in Tumorigenesis

To illustrate the ambiguous role of fibroblasts in tumor progression, we explore different scenarios through a 2-dimensional numerical study. This allows us to compare a fibrotic and a non-fibrotic tumor in the presence or absence of an immune response. Thus, the different growth cases we present are: a tumor without CAFs and T-cells, a tumor with both CAFs and T-cells, a tumor with T-cells but with a low level of CAFs, and a case with CAFs but no efficient T-cells. In the Supplementary Information, we present other scenarios of free growth: a TME in the presence of only NAFs and inefficient T-cells (**Fig. Sup. 1B**), in presence of NAFs (**Fig. Sup. 1C**), and a TME in which only cancer cells are present (**Fig. Sup. 1D**). We first consider a single tumor nest before analyzing the case of two adjacent nests. Therefore, there is only one tumor nest at time $t = 0$ (**Fig. 4A**).

As shown in **Fig. 4E,H** (blue curve), in the case of a free-growing tumor, i.e. in the presence of NAFs and inactive T-cells, cancer cell growth is not hindered by any obstacles, so this is the most severe situation. However, the tumor provokes the formation of a stroma composed of the NAFs and inefficient T-cells. The presence of a stroma plays a role in the cancer cell growth described in **Eq. 1** and leads to a growth that would be less important than in a case without stroma or composed only of NAFs (**Fig. Sup. 1A** → **G**). In fact, the pressure exerted by the stroma increases the mass fraction of cancer cells in the core of the nest, as well as the total mass fraction N at the stroma-nest interface. This phenomenon corresponds to the competition for space and resources between the different species already described in the previous section. In the case of a free cancer growth without stroma, the tumor nest rapidly invades the environment and its surface fraction in the simulation window is almost 40% after reaching $t = 35 T_0$, with a trend that is not yet saturated and an average mass fraction of cancer cells of 25%.

When fibroblasts are activated, they make the environment around the tumor nest fibrotic (**Fig. 4B,C**), with a mass fraction around the nest reaching 20-30%. As explained earlier (in **Section III.1**), the fibers are introduced in our model by an increase of the friction coefficient between the different species and the activated fibroblasts, which is directly related to the fiber concentration. The tumor cells are then trapped behind a barrier with a very high friction which prevents the nest from expanding, and the surface area of the tumor nest decreases to 25% after $35 T_0$, with a fibrotic area reaching 70%, while the average cancer cell decreases to 17% (**Fig. 4E,F,G** cyan curve). The fibroblast population is composed of 30% NAFs and 70% CAFs, as the average fibroblast mass fraction is 26% (**Fig. 4I,K** cyan curve).

At the same time, even in a situation where T-cells are efficient, the barrier precludes T-cells from the tumor (**Fig. 4C**). In the latter case, tumor integrity is maintained and CAFs play a tumor-promoting role, inhibiting the immune response and stabilizing the tumor nest at 7.5% of the domain, with a fibrotic zone of 50% (**Fig. 4E,F,H**, green curve). In this scenario, the average mass fractions are 5% for cancer cells, 10% for CAFs, 10% for NAFs (so 20% for the fibroblast population) and 20% for T-cells. It is interesting to note that this scenario quickly leads to a stable steady state. In contrast, the other simulations take longer times to reach a steady state. This could be due to the fact that the stroma builds up quickly compared to cell death when the cell population is not renewed, and that the steady state corresponds to a small nest that is reached after a short period of growth.

When T-cells are introduced without transformation from NAFs to CAFs (**Fig. 4I** -red curve), the tumor dies as cancer cells are eliminated and the compact nest disappears (**Fig. 4E** -red curve). As the stroma initially builds up due to the presence of a tumor, the fibroblast and T-cell populations slowly relax. However, the tumor stroma takes a long time to retract. This suggests that even if the cancer is cured, the effects on the tissue can be long-lasting. In this case of non-

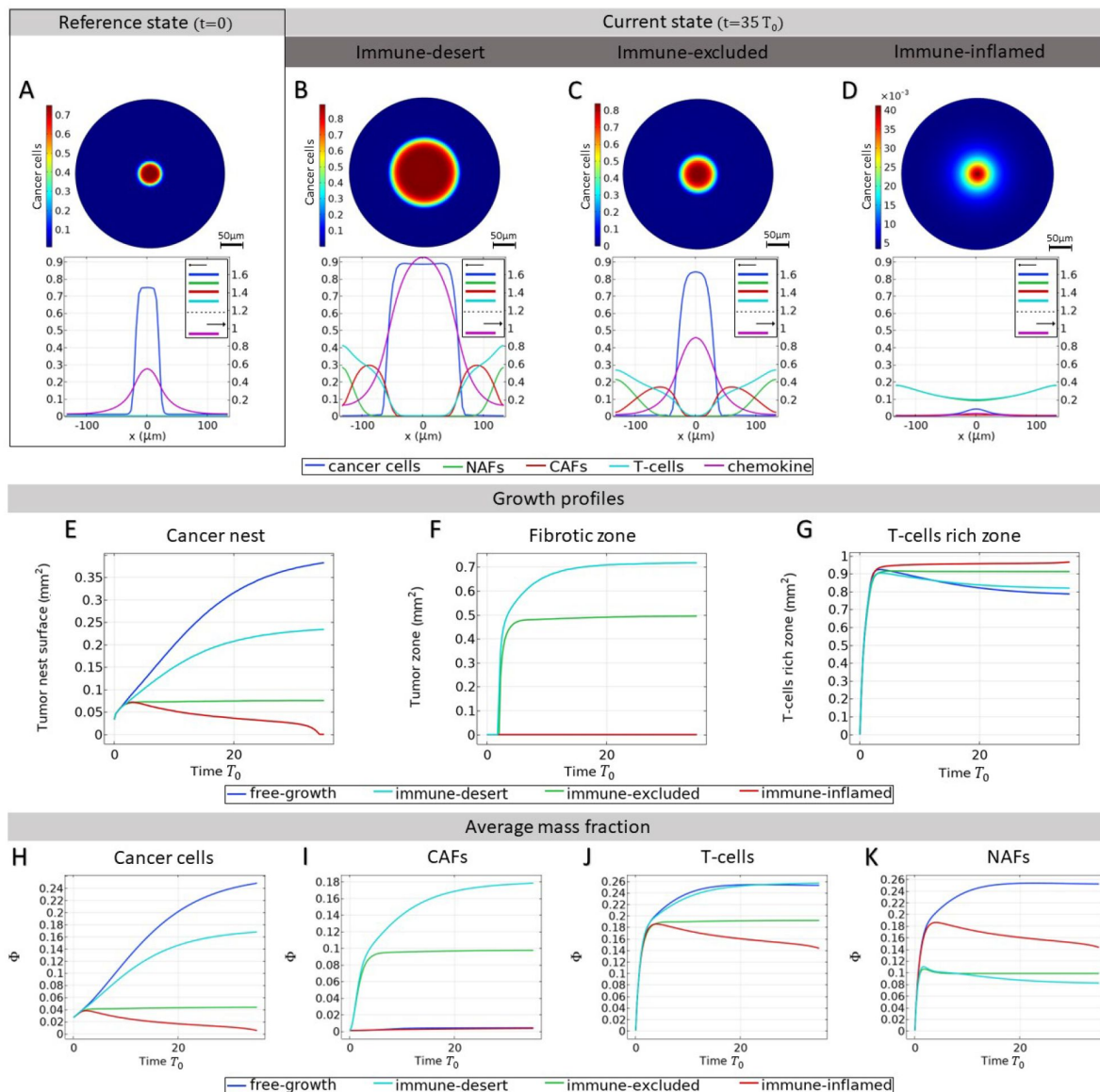


Figure 4

Small solid tumor growth.

We refer here to the different tumor phenotypes described in Fig. 1. **A**: Mass fraction of cancer cells at time $t = 0$ and profile of different mass fractions on a section of the tumor. **B-C-D**: Mass fraction of cancer cells at time $t = 35T_0$ and profile of mass fraction in immunodeficient tumor (**B**), when immune infiltration is inhibited by CAFs (**C**), and in immune-inflamed tumor (**D**). In (**D**), the scale of the colorbar is 10^{-3} the values of (**A**), (**B**), (**C**), since this panel represents the case of efficient T-cells.

E-F-G: Development of different zones. **E**: Surface fraction profile of the tumor nest for different scenarios, calculated as $S/S_{total} = \int dV \delta(C > 0.1)/S_{total}$. **F**: Fibrotic surface fraction profile for different scenarios, calculated as $S/S_{total} = \int dV \delta(CAF > 0.1)/S_{total}$. **G**: T-cell rich area fraction profile for different scenarios, calculated as $S = \int dV \delta(T > 0.1)/S_{total}$. **H**: Cancer cell average mass fractions. **I**: CAF average mass fractions. **J**: T-cell average mass fractions. **K**: NAF average mass fractions.

activation of NAFs, T-cells infiltrate the core of the nest thanks to the attraction of cancer cells (through the parameter λ_{CT}) and allows tumor reduction (**Fig. 4D**). Therefore, their invasion is efficient only in the absence of fibers. (**Fig. 4G**-red curve).

There are many mechanisms able to inhibit the immune system. In addition to the exclusion of T-cells from the tumor and the absence of active feedback, low attractiveness can also reduce the immune response. Indeed, when T-cells are less attracted, for example because chemotaxis is not efficient enough, the tumor is free to expand. In **Fig. 4B**, the cytotoxicity of T-cells is impaired, but their chemotaxis from the boundaries of the domains is not. This leads to an accumulation of inefficient T-cells around the tumor nest.

Next, we analyze the case where two tumor nests are nucleated and interact. We assume that they have the same initial size and mass fraction of cancer cells. In the absence of CAFs, an immune response is triggered and T-cell infiltration can occur. As expected, in the immuneinflamed tumor, the activity of T-cells reduces the ability of the two nests to coalesce by also reducing their mass fraction, leading to very low values of cancer cells (see **Fig. 5C**). When T-cell activity is marginalized by the presence of CAFs, the two nests slowly coalesce to form a single solid tumor nest (see **Fig. 5B**). In this case, growth is actually limited by the tumor-promoting function of CAFs. Although fibroblasts surround the tumor, its growth is enhanced due to the lack of immune cells inside. On the contrary, in the absence of T-cell chemotaxis, the tumor growth is unrestricted and the nests create a larger tumor by also increasing their mass fraction (see **Fig. 5A**). In the latter cases, coalescence leads to anisotropic shapes of the tumor. Although relaxation in our model eventually leads to a round shape, this relaxation is slower for large tumors and even more for fibrotic tumors where friction significantly slows down this process. Patterns associated with coalescence may thus provide insights into the interpretation of anisotropic patterns in tumors. These are also related to the particular geometry of the system, such as the shape of the organ and the location of various blood vessels. In the next paragraph, we provide different directions to better describe this anisotropy.

III.4. Toward anisotropy

To gain insight into the mechanisms leading to anisotropy, we numerically study different cases where the blood vessels are not spatially homogeneously distributed in the vicinity of the tumor nest. Thus, we introduce in our systems localized sources of T-cells and NAF, in the sense of **Fig. 2**, where blood vessels are disk-like patterns in the tissue slice (**Fig. 6A**), or sources corresponding to only a part of the boundaries, as it would be the case for a blood vessel located in the plane of the tissue slice (**Fig. 2**). We note that the anisotropy in the sources can lead to non-trivial shapes of the tumor tissue that are not due to instability but rather to the geometry of the system. These vessels are considered static, as dynamic vessels would require a more accurate modeling of angiogenesis. In the case of localized blood vessels modeled as disks, since the interface between the vessel and the system is smaller than in the case of boundary sources, the parameters driving T-cell population growth and T-cell efficiency in killing cancer cells must be increased in order to eradicate the tumor in the absence of cancer-associated fibroblasts. This highlights a peculiar phenomenon: increasing the density of blood vessels has ambiguous consequences. On the one hand, it allows a more efficient immune response and activate the arrival of T-cells. However, on the other hand, it also provides more nutrients to cancer cells and could ultimately favors metastasis. Therefore, critical processes such as the presence of a fibrous barrier around the tumor may have dual effects on and during tumor evolution.

Furthermore, we introduce a more complete description of the fibrous stroma by adding a nematic order to the matrix and an anisotropic friction with respect to this matrix. A tensor order parameter Q is introduced which characterizes the average orientation of the fibroblasts and fibers and their degree of order (see Appendix B). Indeed, it has been shown that the density of the extracellular matrix is not always sufficient to induce T-cell exclusion [12]. We therefore

Figure 5

Growth process of two small solid tumors.

Density plots showing cancer cell population. We start with two tumor nests placed at the same distance from the center of the domain ($30\mu\text{ m}$), sharing the same mass fraction of cancer cells and we present different scenarios. **A:** Nests coalesce in immune-desert tumor, i.e. in the presence of inefficient T-cells. **B:** Nests interacting in immune-excluded tumor, with chemotactic T-cells and CAFs. **C:** Nests coalesce in immune-inflamed tumor.

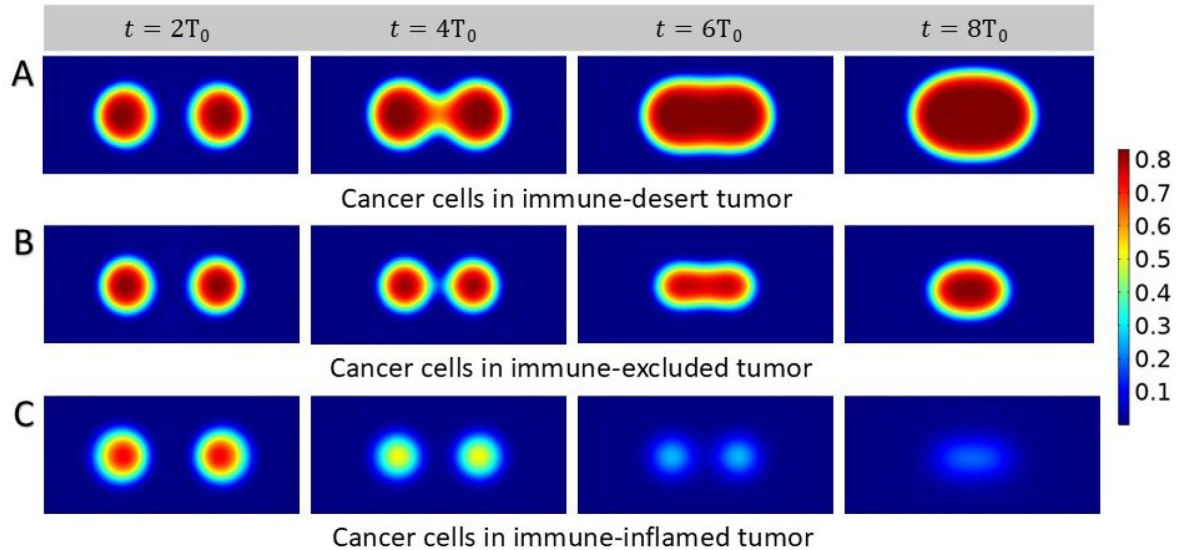
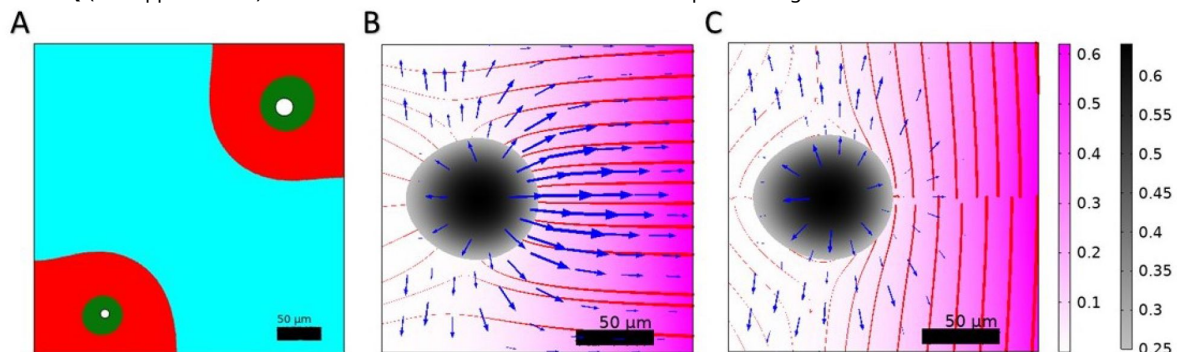


Figure 6

Anisotropy.

Following biological observation reported in [Fig. 2](#), we here show different cases for describing anisotropy. **A:** Example of the tumor microenvironment in the presence of two blood vessels (in white). The tumor nest is in cyan ($C > F_A + F_{NA} + T$), the cancerous stroma in red ($F_A > F_{NA} + T$) and the healthy stroma in green ($F_{NA} + T > F_A$). The tumor shape is dictated by the localization of the blood vessels from which T-cells and NAFs are issued. **B-C:** Tumor behavior in the presence of normal (**B**) or orthonormal (**C**) fibers. The tumor fraction is indicated in colors from black to grey which we limit for $C > 0.25$ to better visualize of the contour of the nest. The flux $C \mathbf{v}_C$ is indicated with blue arrows whose thickness is proportional to the magnitude of the flux. The nematic order is shown with red lines whose thickness is proportional to the determinant of the matrix $\mathbf{Q}$ (see Appendix B.1) and the fibroblast fraction is indicated with a pink color gradient.



provide the CAFs with an orientation, that is the result of their coupling with the matrix they deposit [76, 77]. This long range order has been shown to induce an anisotropic friction [78] that may be related to immune cell exclusion [79]. At the same time, matrix orientation strongly influences metastasis and tumor growth by providing directional cues. Thus, there is a strong difference in the outcome of a tumor with fibers normal to the tumor surface and one with fibers along the tumor, both at the level of immune response and cancer escape. In **Fig. 6B,C** we show the nematic order of the matrix and the flux field of the cancer cells in the case of orthonormal and normal orientation with respect to the tumor nest boundary, and due to the influx of fibroblasts from the right side of the system. The flux of cancer cells is much higher in the case of normal orientation of the fibers than in the case of parallel orientation. In conclusion, the description of T-cell exclusion by the sole density would be adapted to a case with fibers orientation aligned with the cancer nest surface, but not in the case with different orientations.

Further investigations may therefore introduce these two elements to our model: a nematic order in the CAF layer, whose orientation is determined by coupling with the orientation of the interfaces, and an anisotropic friction, which is higher in the direction normal to the nematic matrix order than in the orientation of the matrix. Note that active stresses ($\sigma = -\zeta Q$ with ζ the activity) and couplings with the proliferation can develop in interaction with this nematic order, as explained and modeled in a previous article by some of us [61], and a precise description of the nematic order may require to describe separately the fibroblasts and the matrix [76, 77, 78].

IV. Discussion

Our work provides a physical model to quantify the role of the immune system in human lung tumors. This is in line with different similar mathematical models, that study through this lens the inhibition/activation of the immune system by cancer cells either by the mean of nonlinear compartment models resembling our dynamical system, for instance regarding macrophage enrolment and cytokine signaling [80, 81], or mixture models [82]. We thus connect the two approaches in order to rigorously derive the parameters of the model and derive insights from both. In our article, the focus is on early tumor growth prior to angiogenesis and the development of its stroma rich in T-cells and fibroblasts, the activation of non-activated fibroblasts into cancer-associated fibroblasts and the marginalization of T-cells from the tumor nest. Other immune cells, such as macrophages, could complete this study in a further investigation.

After analyzing the data from the literature and from patients with LUSC or LUAD pathologies, we propose a physical model, both theoretical and numerical, for the interactions between cancer cells, T-cells and fibroblasts during tumor progression and their different roles. In particular, fibroblasts are introduced in the model in their inactive and active states (NAFs and CAFs). The modeling involves two steps: firstly, a dynamic study aimed at elucidating the key factors that can characterize the properties of the micro-environment on which the ability of T-cells to inhibit tumor growth depends. Scenarios scaled by only two parameters control the dynamics and evaluate the aggressiveness of the tumor. In particular, we have established the spatial organization of the different cell types inside and outside the tumor core. The model is based on the continuous mixture model derived from Onsager's variational principle. Simulations were performed with the FEM software COMSOL Multiphysics in a two-dimensional framework. This provided a complete description of the tumor morphology and composition. In fact, we explored different scenarios to fully appreciate the role of all cell types involved. The results confirmed the experimental evidence that CAFs play a dual role in promoting and suppressing cancer cell proliferation. First, they alter the tumor microenvironment with a fiber barrier that reduces the motility and activity of T-cells, thereby promoting cancer cell growth. At the same time, this growth is limited and results to be lower than in the absence of active fibroblasts. Moreover, we

have also shown that although the different parameters can take a wide range of values and continuously change the results of the numerical study, different scenarios can be drawn by looking only at the orders of magnitude.

In this study, we did not consider the fibers of the extracellular matrix as a component *per se*, but we consider them through the friction increase. We also provided the basics and started to describe the consequences of an anisotropic friction in relation to a nematic fiber orientation (see Appendix B.2). The same formalism could also integrate non trivial active stresses as done in Ref. [61].

Our physical model aims to limit the number of parameters as much as possible. This may seem far-fetched given the complexity of the biological system, especially *in vivo*. However, there are several reasons for this limitation. First, determining the range of values for these parameters is a complex task in the absence of direct measurement. The strategy we used in Section II is to isolate each process corresponding to each parameter and thus construct the model step by step. In Section III we used both this latter strategy and the spatial structure of the mixing. In addition, the different combinations of the parameter values provide a wide variety of scenarios and it would be sufficient to modify these values to model new chemical entering the system. Limiting the complexity of the different processes allows us to focus on other aspects of the problem, such as the time evolution and the spatial structure, as well as a more precise quantitative study. In the same spirit, we numerically report in Section III a 2-dimensional system without substrate, which yields results that we can consider similar as a 3-dimensional system with an invariance in the third dimension.

Identifying the different scenarios that can occur in the TME and their likelihood is critical to predict the outcome of therapy [60]. Indeed, both pharmacodynamics and pharmacokinetics are highly dependent on the spatial structure and composition at the tumor site. In particular, the dynamic of arrival of the T-cell population is over simplified here via chemotaxis and discard auto-chemotaxis at the origin of a swarming displacement of an increased number of cytotoxic T-cells eventually, CAR-T cells [83]. This mechanism, rare in the solid-tumor case, is sequential in time leading to a competition between killing efficiency and cancerous cell repair [84, 85] which will depend on the cancer type considered. Classification and quantitative characterization of the various outcomes can also help to monitor the treatment in real time according to the evolution of the TME.

In conclusion, this numerical investigation may help to understand the limitations of the immune system in the face of solid tumor growth. Understanding quantitatively how immune cells are excluded from the tumor nest may be helpful in the drug design of T-cell-based therapies. Conversely, the numerical reproduction of various processes that can be observed *in vivo* is an important step in the context of personalized medicine. A next step in the theoretical and numerical investigation would therefore be to introduce drug molecules into the framework we have presented. Indeed extending our model to follow the tumor evolution under treatment and the associated internal dynamics from the early alveolar stage of the lung by incorporating the actual three-dimensional environment would also be a major step forward.

Supporting information

In this appendix, we show cases of free growth but in simple stroma: the case of a stroma composed by NAFs and inefficient T-cells (Fig. Sup. 1B), a stroma composed by NAFs only (Fig. Sup. 1C), and the case without stroma (Fig. Sup. 1D), the tumor nest surface fraction in the simulation window for the different scenarios (Fig. Sup. 1E), and the average mass fraction of the cancer cells and the NAFs (Fig. Sup. 1F,G). The values of the parameters depending on the scenario are shown in Table. Sup 1.

Parameter	$\tau_T (\mu\text{m.d}^{-1})$	$\tau_{NAF} (\mu\text{m.d}^{-1})$
C-c+NAFs+inefficient T-cells	2.5×10^{-3}	2.5×10^{-3}
C-c+NAFs	2.5×10^{-6}	2.5×10^{-3}
C-c	2.5×10^{-6}	2.5×10^{-6}

Table Sup 1

Parameter values varying depending on the scenario in Fig. 1 [↗](#).

Besides, we show the evolution of the tumor in the case with efficient T-cells and NAFs without CAFs (**Fig. Sup. 1H** [↗](#)).

We also obtain different configurations by assuming deformed domains in order to study the anisotropic response of the tumor stroma (**Fig. Sup. 2A,B** [↗](#)).

In particular, the results obtained differ from those presented above by introducing T-cells and NAFs only from the upper side of the boundaries, this assuming the possibility that T-cells are able to reach the nest from different parts. Finally, different regions have been highlighted: in *light blue* the cancer cells nest, in *red* the barrier of activated fibroblasts and in *green* the healthy stroma composed of NAFs and T-cells.

Data and materials availability

All data are available in the main text, in the figures and in the tables. Comsol Multiphysics files and raw data used in this work are available upon request.

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Additional information

Author contributions

M.BA and J.A conceived and designed the project, built the physical model and performed the theoretical analysis, assisted by M.F and C.B. H.S and P.S provided the biological data and images. C.B and J.A performed the numerical simulations. M.BA, J.A, and C.B wrote the manuscript with contributions from P.S, H.S, and M.F.

Appendix A. Dynamical System

In this section, we detail the calculations referred to in the dynamical system part of the article (**Section II** [↗](#)). More precisely, we demonstrate the scaling of the various parameters in the case where activated fibroblasts prevent T cells from killing the cancer cells. With the found scaling, we show that the stationary solutions of this dynamical system are the only stable solutions.

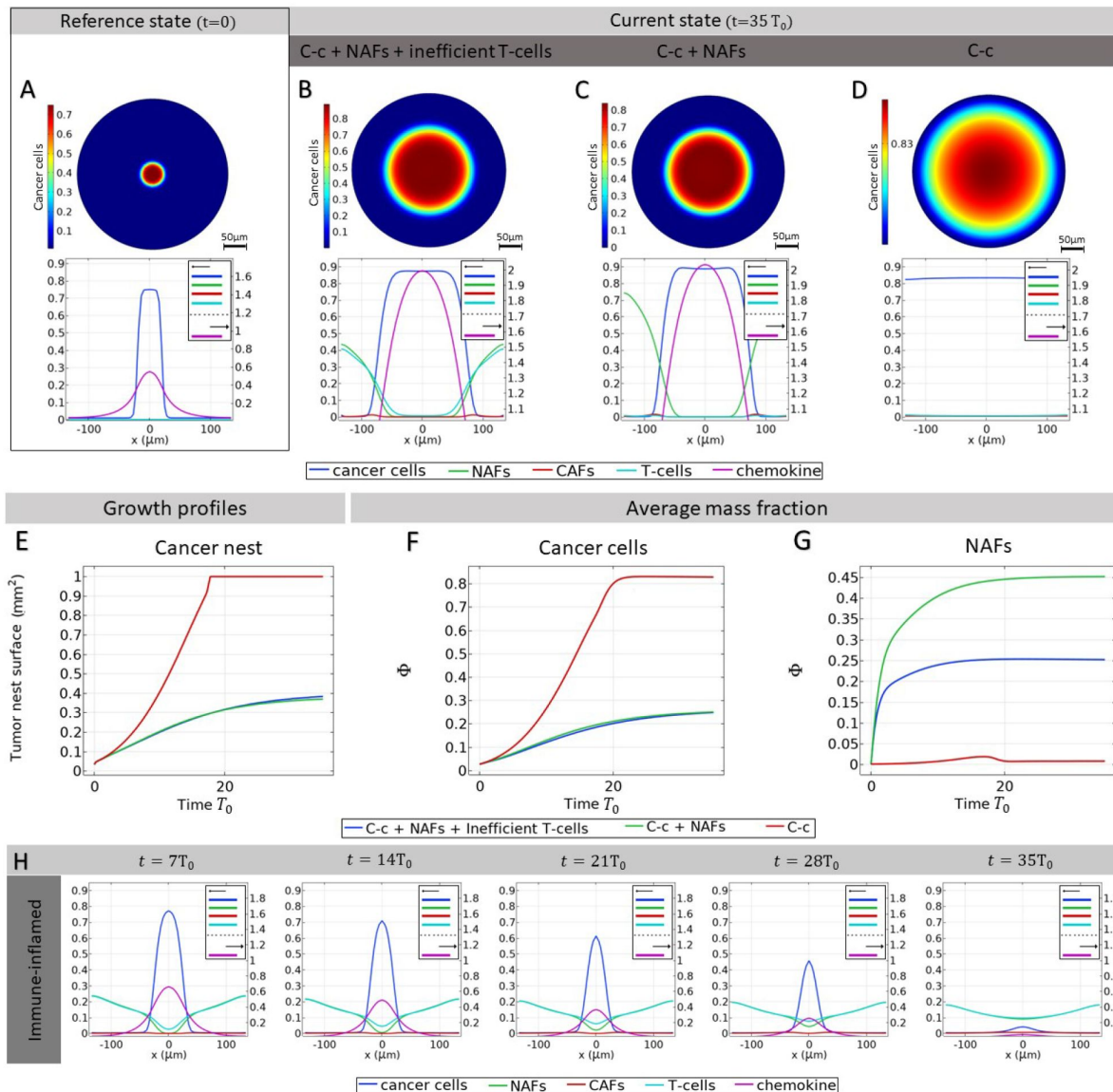


Figure Sup 1

Free growth of a tumor nest.

A-G: Cancer population density plots for different scenarios: Cancer cells (C-c)+NAFs+Inefficient T-cells, Cancer cells+NAFs, Cancer cells alone. **A:** Mass fraction of cancer cells at time $t = 0$ and profile of different mass fractions on a section of the tumor. **B:** Mass fraction of cancer cells at time $t = 35T_0$ and mass fraction profile for a case of a stroma composed by C-c, NAFs, and inefficient T-cells. **C:** Mass fraction of cancer cells at time $t = 35T_0$ and profile of the different mass fractions on a cut of the tumor for a case of a stroma composed by C-c and NAFs. **D:** Mass fraction of cancer cells at time $t = 35T_0$ and mass fraction profile for a case with no stroma in the tumor microenvironment. **E:** Surface fraction of the tumor nest in the different scenarios. **F:** Average mass fraction of the cancer cells in the different scenarios. **G:** Average mass fraction of the NAFs in the different scenarios. **H:** Time evolution of the mass fraction profile of the cut of the simulation window in the case of a tumor in the presence of efficient T-cells and NAFs only, without T-cells.

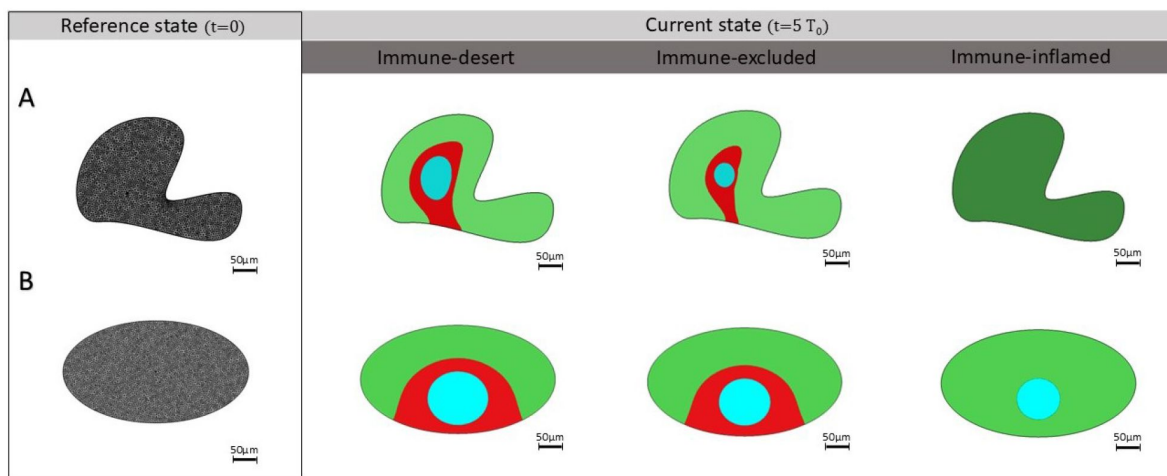


Figure Sup 2

Different tumor phenotypes in non-spherical domains.

Deformed shapes result in anisotropic configurations by forcing NAFs and T-cells to enter the domain from only one side of the boundary. Different colors indicate separated regions where the nest (in light blue) is confined by the boundary of CAFs (in red) within a healthy stroma (in green). **A:** Immune-inflamed case leads to disappearance cancer cells at $t = 5T_0$. **B:** Elliptical area leads to slower dynamics.

First, we recall the complete system of equations describing the dynamics of the tumor microenvironment. C , T , F_A , F_{NA} represent the cancer cells, T cells, active fibroblasts, and inactive fibroblasts, respectively.

$$\begin{cases} \frac{dC}{dt} = C - \frac{\delta_{CT}CT}{1+\delta_{TF}F_A} - \delta_C C \mathcal{S}, \\ \frac{dT}{dt} = \alpha_{TC}C - \delta_T T \mathcal{S}, \\ \frac{dF_{NA}}{dt} = \alpha_{NA} + \alpha_{NA,C}C - K_A F_{NA}C - \delta_{NA} F_{NA} \mathcal{S}, \\ \frac{dF_A}{dt} = K_A F_{NA}C - \delta_A F_A \mathcal{S}, \end{cases} \quad (\text{B.1})$$

where δ_{CT} is the killing rate of cancer cells by T cells in the absence of active fibroblasts, δ_{TF} is the inhibition constant of T cells by the active fibroblasts, α_{TC} is the attraction rate of T cells to the tumor, α_{NA} is the attraction rate of inactive fibroblasts in the healthy tissue, $\alpha_{NA,C}$ is the attraction rate of non-active fibroblasts in the presence of cancer cells, K_A is the plasticity constant between active and non-active fibroblasts, and δ_C , δ_T , δ_{NA} , δ_A are the death rates due to space and resource constraints of the different species. For simplicity, in this article we consider $\delta_C = \delta_T = \delta_{NA} = \delta_A = \delta$. We also define the small parameter $\epsilon \ll 1$.

Appendix A.1. Interaction between T-cells and cancer cells

We examine the interaction between T cells and cancer cells in the nest in the absence of fibroblasts $F_{NA} = F_A = 0$ as described in [Section II.2.2](#). This case would lead to the tumor eradication. We look for parameters that lead to significant growth inhibition such that $C \sim \epsilon$ at equilibrium.

The equilibrium equations read:

$$\begin{cases} \frac{dC}{dt} = 0 = C - \delta_{CT}CT - \delta C(C + T), \\ \frac{dT}{dt} = 0 = \alpha_{TC}C - \delta T(C + T). \end{cases} \quad (\text{B.2})$$

Three pairs of solutions exist:

$$\{C, T\} = \left\{ \begin{array}{cc} 0 & , & 0 \\ -\frac{\alpha_{TC}\delta^2 + 2\alpha_{TC}\delta\delta_{CT} - (\delta + \delta_{CT})\sqrt{\Delta} + \alpha_{TC}\delta_{CT}^2 + \delta^2 - \delta\delta_{CT}}{2\delta^2\delta_{CT}} & , & \frac{-\sqrt{\Delta} + \alpha_{TC}\delta + \alpha_{TC}\delta_{CT} + \delta}{2\delta\delta_{CT}} \\ -\frac{\alpha_{TC}\delta^2 + 2\alpha_{TC}\delta\delta_{CT} + (\delta + \delta_{CT})\sqrt{\Delta} + \alpha_{TC}\delta_{CT}^2 + \delta^2 - \delta\delta_{CT}}{2\delta^2\delta_{CT}} & , & \frac{\sqrt{\Delta} + \alpha_{TC}\delta + \alpha_{TC}\delta_{CT} + \delta}{2\delta\delta_{CT}} \end{array} \right\}, \quad (\text{B.3})$$

with $\Delta = (\alpha_{TC}\delta + \alpha_{TC}\delta_{CT} + \delta)^2 - 4\alpha_{TC}\delta\delta_{CT}$.

Assuming $\delta_{CT} = \delta\epsilon^{-1}$, $\alpha_{TC} = a_0\epsilon$, with $\epsilon \ll 1$, $a_0 \sim 1$, the solutions, [eq.\(B.3\)](#) simplify:

$$\{C, T\} = \left\{ \begin{array}{cc} 0 & , & 0 \\ \frac{\epsilon}{(a_0-1)\delta} & , & \frac{\epsilon}{\delta} \\ \frac{1-a_0}{\delta} + \mathcal{O}(\epsilon) & , & \frac{a_0\epsilon}{\delta} \end{array} \right\}. \quad (\text{B.4})$$

The only solution leading to the tumor eradication is $\{C, T\} = \{\frac{\epsilon}{(a_0-1)\delta}, \frac{\epsilon}{\delta}\}$, and $a_0 > 1$.

We now check the linear stability of the different solutions by evaluating the Jacobian eigenvalues. A solution is stable if all its eigenvalues have a negative real part [86]. The solution found is also the only stable solution provided that $a_0 > 1$. Here, the different eigenvalues of the Jacobian matrix are for the different solutions:

$$\{\omega_1, \omega_2\} = \left\{ \begin{array}{cc} 1 & , \quad 0 \\ -\frac{a_0\epsilon + \sqrt{\epsilon} \sqrt{a_0(2a_0-1)\epsilon - (a_0-1)^2}}{a_0-1} & , \quad \frac{\sqrt{\epsilon} \sqrt{a_0(2a_0-1)\epsilon - (a_0-1)^2} - a_0\epsilon}{a_0-1} \\ \frac{a_0\epsilon}{a_0-1} + a_0 - 1 & , \quad a_0 \left(\frac{a_0\epsilon}{a_0-1} + \epsilon + 1 \right) - 1 \end{array} \right\}. \quad (\text{B.5})$$

The second solution with $a_0 > 1$ is the only stable solution.

Appendix A.2. Role of activated fibroblasts on T-cells

We isolate a subsystem composed of cancer cells, T-cells and active fibroblasts, as in Section II.2.2, in order to determine the inhibition rate δ_{TF} , responsible for the marginalization of the T-cells and subsequently for the increase of cancer cells at a fixed volume fraction of fibroblasts. We first assume that F_A is constant. The equilibrium equations read:

$$\begin{cases} \frac{dC}{dt} = 0 = f_0 C - \Delta_{CT} C T - \delta C (C + T), \\ \frac{dT}{dt} = 0 = \alpha_{TC} C - \Delta_F T - \delta T (C + T), \end{cases}$$

where $f_0 = 1 - \delta F_A$, $\Delta_{CT} = \delta_{CT} (1 + \delta_{TF} F_A)^{-1}$ and $\Delta_F = \delta F_A$.

The different pairs of solutions read:

$$\{C, T\} = \left\{ \begin{array}{cc} 0 & , \quad 0 \\ 0 & , \quad -\frac{\Delta_F}{\delta} \\ \frac{\delta f_0 (\Delta_{CT} - \delta) - (\Delta_{CT} + \delta) \left(\alpha_{TC} (\Delta_{CT} + \delta) + \delta \Delta_F - \sqrt{(\alpha_{TC} (\Delta_{CT} + \delta) + \delta \Delta_F + \delta f_0)^2 - 4 \delta f_0 \alpha_{TC} \Delta_{CT}} \right)}{2 \delta^2 \Delta_{CT}} & , \quad \frac{-\sqrt{(\alpha_{TC} (\Delta_{CT} + \delta) + \delta \Delta_F + \delta f_0)^2 - 4 \delta f_0 \alpha_{TC} \Delta_{CT}} + \delta \alpha_{TC} + \alpha_{TC} \Delta_{CT} + \delta \Delta_F + \delta f_0}{2 \delta \Delta_{CT}} \\ \frac{\delta f_0 (\Delta_{CT} - \delta) - (\Delta_{CT} + \delta) \left(\alpha_{TC} (\Delta_{CT} + \delta) + \delta \Delta_F + \sqrt{(\alpha_{TC} (\Delta_{CT} + \delta) + \delta \Delta_F + \delta f_0)^2 - 4 \delta f_0 \alpha_{TC} \Delta_{CT}} \right)}{2 \delta^2 \Delta_{CT}} & , \quad \frac{\sqrt{(\alpha_{TC} (\Delta_{CT} + \delta) + \delta \Delta_F + \delta f_0)^2 - 4 \delta f_0 \alpha_{TC} \Delta_{CT}} + \delta \alpha_{TC} + \alpha_{TC} \Delta_{CT} + \delta \Delta_F + \delta f_0}{2 \delta \Delta_{CT}} \end{array} \right\}. \quad (\text{B.6})$$

We assume $\delta_{CT} = \delta \epsilon^{-1}$, $\alpha_{TC} = a_0 \epsilon$, $\delta_{TF} = d_0 \delta \epsilon^{-1}$, as justified in the main text (Section II.2.2). We obtain the following solution pairs:

$$\{C, T\} = \left\{ \begin{array}{cc} 0 & , \quad 0 \\ 0 & , \quad -\frac{f_a}{\delta} \\ \frac{f_0}{\delta} + \mathcal{O}(\epsilon) & , \quad \frac{a_0 f_0}{\delta (f_0 + f_a)} \epsilon \\ -\frac{f_a (d_0 f_0 + d_0 f_a + 1)}{\delta} + \mathcal{O}(\epsilon) & , \quad \frac{d_0 f_a (f_0 + f_a)}{\delta} + \mathcal{O}(\epsilon) \end{array} \right\}. \quad (\text{B.7})$$

The only physical solution is the third pair: $\{C, T\} = \left\{ \frac{f_0}{\delta} + \mathcal{O}(\epsilon), \frac{a_0 f_0}{\delta (f_0 + f_a)} \epsilon \right\}$.

The eigenvalues of the Jacobian matrix read:

$$\{\omega_1, \omega_2\} = \left\{ \begin{array}{cc} -f_a & , \quad f_0 \\ f_a & , \quad \left(\frac{1}{d_0} + f_0 + f_a\right) \\ -(f_0 + f_a) & , \quad -f_0 \\ \frac{f_a - i\sqrt{f_a}\sqrt{4f_0(d_0f_0 + 2d_0f_a + 1) + f_a(4d_0f_a + 3)}}{2} & , \quad \frac{f_a + i\sqrt{f_a}\sqrt{4f_0(d_0f_0 + 2d_0f_a + 1) + f_a(4d_0f_a + 3)}}{2} \end{array} \right\}, \quad (\text{B.8})$$

and the third solution is the only stable one.

Appendix A.3. Fibroblast plasticity

We consider the system described in Section II.2.4, where the non-active fibroblasts population is maintained at a fraction $F_{NA} = f_n \delta^{-1}$ where $f_n < 1$ is a constant.

We write the equilibrium equations, where we replace $\delta_{CT} = \delta \epsilon^{-1}$, $\alpha_{TC} = a_0 \epsilon$, $\delta_{TF} = d_0 \delta \epsilon^{-1}$:

$$\begin{cases} \frac{dC}{dt} = 0 = C - \frac{\delta \epsilon^{-1} CT}{1 + d_0 \delta \epsilon^{-1} F_A} - \delta C \mathcal{S}, \\ \frac{dT}{dt} = 0 = a_0 \epsilon C - \delta T \mathcal{S}, \\ \frac{dF_A}{dt} = 0 = K_A f_n C / \delta - \delta F_A \mathcal{S}, \end{cases} \quad (\text{B.9})$$

where $\mathcal{S} = C + T + F_A + F_{NA}$.

Equating $C = \delta T \mathcal{S} / (a_0 \epsilon)$ and $C = \delta^2 F_A \mathcal{S} / (K_A f_n)$ leads to $F_A = K_A f_n T / (\delta a_0 \epsilon)$. The system rewrites:

$$\begin{cases} 0 = (1 - f_n)C - \left(\frac{\delta \epsilon^{-1}}{1 + d_0 \delta \epsilon^{-2} K_A f_n T / a_0} + \frac{K_A f_n}{a_0 \epsilon}\right)CT - \delta C(C + T), \\ 0 = a_0 \epsilon C - f_n T - \frac{K_A f_n}{a_0 \epsilon} T^2 - \delta T(C + T). \end{cases} \quad (\text{B.10})$$

Since the active fibroblasts should prevent T-cells to kill cancer cells, but without reducing the T-cell population, we write $C \sim \delta^{-1}$ and $T \sim \epsilon \delta^{-1}$. In addition, the term related to the killing of cancer-cells by T-cells should be of order ϵ : $\frac{\delta \epsilon^{-1}}{1 + d_0 \delta \epsilon^{-2} K_A f_n T / a_0} T \epsilon \sim \epsilon$. This leads to: $K_A \sim \delta$.

We now look for the solutions of the dynamical system, as expansions in ϵ . Obvious solutions are: $C = 0$, $T = 0$ and $C = 0$, $T = -f_n \epsilon / (\delta \epsilon + K_A f_n / a_0)$. We look for the other solutions in the form: $C = c_0 + c_1 \epsilon$, $T = t_0 + t_1 \epsilon$. The order $1/\epsilon$ in the dynamical system lead to $t_0 = 0$. The orders 1 and ϵ in the dynamical system leads to:

$$\begin{aligned} 0 &= (1 - f_n) - \frac{K_A f_n}{a_0} t_1 - \delta c_0 \\ 0 &= a_0 c_0 - f_n t_1 - \frac{K_A f_n}{a_0} t_1^2 - \delta t_1 c_0 \end{aligned} \quad (\text{B.11})$$

Finally the different solutions are:

$$\{C, T\} = \left\{ \begin{array}{cc} 0 & , \quad 0 \\ 0 & , \quad -\frac{a_0 \epsilon}{K_a} + \mathcal{O}(\epsilon^2) \\ \frac{1-f_n}{K_a f_n + \delta} + \mathcal{O}(\epsilon) & , \quad \frac{a_0 \epsilon (1-f_n)}{K_a f_n + \delta} + \mathcal{O}(\epsilon^2) \end{array} \right\} \quad (\text{B.12})$$

with only one physical solution apart from the disappearance of both T-cells and cancer cells.

We now check the stability of the different solutions. The eigenvalues for the growth rates are:

$$\{\omega_1, \omega_2\} = \left\{ \begin{array}{cc} 1-f_n & , \quad -\epsilon f_n \\ 1 & , \quad \epsilon f_n \\ -\frac{\epsilon(K_a f_n + \delta)}{\delta} & , \quad -\frac{\delta(1-f_n)}{K_a f_n + \delta} + \mathcal{O}(\epsilon) \end{array} \right\} \quad (\text{B.13})$$

and the solution $C = \frac{1-f_n}{K_a f_n + \delta}$, $T = \frac{a_0 \epsilon (1-f_n)}{K_a f_n + \delta}$, $F_A = \frac{K_a f_n (1-f_n)}{\delta(K_a f_n + \delta)}$ is the only stable solution.

Appendix A.4. Non active fibroblasts attraction to tumor

Last, we demonstrate the results of [Section II.2.5](#). In order to analyse non active fibroblasts to tumor, we now consider the full dynamical system:

$$\begin{cases} \frac{dC}{dt} = 0 = C - \frac{\delta \epsilon^{-1} C T}{1 + d_0 \delta \epsilon^{-1} F_A} - \delta C \mathcal{S}, \\ \frac{dT}{dt} = 0 = a_0 \epsilon C - \delta T \mathcal{S}, \\ \frac{dF_{NA}}{dt} = 0 = \delta \epsilon^4 + \alpha_{NA,C} C - k \delta F_{NA} C - \delta F_{NA} \mathcal{S}, \\ \frac{dF_A}{dt} = 0 = k \delta F_{NA} C - \delta F_A \mathcal{S}. \end{cases} \quad (\text{B.14})$$

We look for solutions of the type: $\{C, T, F_{NA}, F_A\} = \delta^{-1} \{c_0 + \mathcal{O}(\epsilon), t_0 \epsilon + \mathcal{O}(\epsilon^2), f_{NA} + \mathcal{O}(\epsilon), f_A + \mathcal{O}(\epsilon)\}$ for the case of fibroblasts preventing T-cells getting access to the tumor.

The dynamical system becomes to the first order in ϵ :

$$\begin{cases} 0 = 1 - \frac{t_0}{1 + d_0 \epsilon^{-1} f_A} - (c_0 + t_0 \epsilon + f_{NA} + f_A), \\ 0 = a_0 c_0 - t_0 (c_0 + t_0 \epsilon + f_{NA} + f_A), \\ 0 = \alpha_{NA,C} c_0 - k f_{NA} c_0 - f_{NA} (c_0 + t_0 \epsilon + f_{NA} + f_A), \\ 0 = k f_{NA} c_0 - f_A (c_0 + t_0 \epsilon + f_{NA} + f_A). \end{cases} \quad (\text{B.15})$$

In order for f_A, f_{NA} to be of order 0 in ϵ , $\alpha_{NA,C}$ must also be of order 0.

At the lowest order in ϵ we obtain:

$$\begin{cases} 0 = 1 - (c_0 + f_{NA} + f_A), \\ 0 = a_0 c_0 - t_0 (c_0 + f_{NA} + f_A), \\ 0 = \alpha_{NA,C} c_0 - k f_{NA} c_0 - f_{NA} (c_0 + f_{NA} + f_A), \\ 0 = k f_{NA} c_0 - f_A (c_0 + f_{NA} + f_A). \end{cases} \quad (\text{B.16})$$

The solutions of the full system write:

$$\{C, T, F_A, F_{NA}\} = \left\{ \begin{array}{cccc} 0 & , & t\epsilon & , & -f_{NA} & , & f_{NA} \\ \frac{1}{\alpha_{NA,C}+1} & , & \frac{a_0}{\alpha_{NA,C}+1} & , & \frac{\alpha_{NA,C}k}{(\alpha_{NA,C}+1)(\alpha_{NA,C}+k+1)} & , & \frac{\alpha_{NA,C}}{\alpha_{NA,C}+k+1} \end{array} \right\}. \quad (\text{B.17})$$

Only the second solution is physical.

We now check the stability of those solutions. The linear stability analysis yields the eigenvalues for the stability rates:

$$\{\omega_1, \omega_2, \omega_3, \omega_4\} = \left\{ \begin{array}{cccc} 1 & , & 0 & , & 0 & , & 0 \\ -1 & , & -1 & , & -\epsilon & , & -\frac{\alpha_{NA,C}+k+1}{\alpha_{NA,C}+1} \end{array} \right\}. \quad (\text{B.18})$$

The only stable solution is the second solution where all cell fractions are above 0 and below 1.

Appendix B. Model for an anisotropic friction with the matrix

Appendix B.1. Nematic order

We introduce a refined model for the fiber role in the tumor microenvironment. We first add a nematic order to the CAF population since its structure involves a more or less disordered nets of fibers. This nematic order is described by a traceless nematic tensor $\mathbf{Q}$. We recall that the nematic tensor may be written as:

$$\mathbf{Q} = \frac{S}{2} \begin{pmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{pmatrix} \quad (\text{C.1})$$

where S is the amplitude of the anisotropy, and θ the angle for the nematic orientation. Besides, $\mathbf{Q}$ is determined by the minimization of the following free-energy:

$$\mathcal{F}_{nem} = \int d\Omega \left\{ \alpha \mathbf{Q} : \mathbf{Q} + \beta (\mathbf{Q} : \mathbf{Q})^2 + \frac{K}{2} (\nabla \mathbf{Q})^2 + \gamma \mathbf{Q} : \Phi \right\} \quad (\text{C.2})$$

where $\alpha = -F_A$, and Φ indicates the orientation of the tumor nest boundary: $\Phi_{ij} = \frac{1}{2}((\partial_i C)(\partial_j C) - 2(\partial_k C)^2)$. The orientation of the fibers with regards to the tumor nest is controlled by the parameter γ . In the spirit of our work, we assume that fibers orient fast compared to the growth times, so that we consider the following equation:

$$\frac{\delta \mathcal{F}_{nem}}{\delta Q} = 0 \quad (C.3)$$

which allows to evaluate Q .

Appendix B.2. Anisotropic friction

We consider an anisotropic friction between the different components and the CAFs in the dissipation function.

$$\mathcal{W}_{anis} = \int d\Omega \left\{ \frac{\xi_{\parallel}}{2} (\mathbf{v}_X^{\parallel} - \mathbf{v}_{FA}^{\parallel})^2 + \frac{\xi_{\perp}}{2} (\mathbf{v}_X^{\perp} - \mathbf{v}_{FA}^{\perp})^2 \right\} \quad (C.4)$$

where $\mathbf{v}_X^{\parallel}$ is the velocity component along the unit vector of the nematic order orientation of the matrix while X represents one component of the mixture $(C, F_A, F_{NA}, T: \mathbf{v}_X^{\parallel} = (\mathbf{v}_X \cdot \mathbf{e}_{\parallel}) \mathbf{e}_{\parallel})$. In the same way, $\mathbf{v}_X^{\perp} = \mathbf{v}_X - \mathbf{v}_X^{\parallel}$. after some algebra, this leads to the nematic friction tensor that depends only on the tensor Q . More justifications can be found in Ref. [78 [↗](#)].

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Reviewer #1 (Public review):

The authors present an important work where they model some of the complex interactions between immune cells, fibroblasts and cancer cells. The model takes into account the increased ECM production of cancer-associated fibroblasts. These fibres trap the cancer but also protect it from immune system cells. In this way, these fibroblasts' actions both promote and hinder cancer growth. By exploring different scenarios, the authors can model different cancer fates depending on the parameters regulating cancer cells, immune system cells and fibroblasts. In this way, the model explores non-trivial scenarios. An important weakness of this study is that, though it is inspired by NSCLC tumors, it is still far from modelling tumor lesions with morphologies similar to NSCLC tumors and does not explore the formation of ramified tumors. In this way, is a general model and it is challenging how it can be adapted to simulate more realistic tumor morphologies.

Comments on revisions:

The authors have improved the manuscript and addressed my concerns.

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Reviewer #2 (Public review):**Summary:**

The authors develop a computational model (and a simplified version thereof) to treat an extremely important issue regarding tumor growth. Specifically, it has been argued that fibroblasts have the ability to support tumor growth by creating physical conditions in the tumor microenvironment that prevent the relevant immune cells from entering into contact with, and ultimately killing, the cancer cells. This inhibition is referred to as immune exclusion. The computational approach follows standard procedures in the formulation of models for mixtures of different material species, adapted to the problem at hand by making a variety of assumptions as to the activity of different types of fibroblasts, namely "normal" versus "cancer-associated". The model itself is relatively complex, but the authors do a convincing job of analyzing possible behaviors and attempting to relate these to experimental observations.

Strengths:

As mentioned, the authors do an excellent job of analyzing the behavior of their model both in its full form (which includes spatial variation of the concentrations of the different cellular species) and in its simplified mean field form. The model itself is formulated based on established physical principles, although the extent to which some of these principles apply to active biological systems is perhaps debatable (see Weaknesses). The results of the model do indeed offer some significant insights into the critical factors which determine how fibroblasts might affect tumor growth; these insights could lead to new experimental ways of unraveling these complex sets of issues and enhancing immunotherapy. In this revised version, the authors have properly placed this work within the general context of other research on modeling the tumor-immune ecology.

Weaknesses:

Models of the form being studied here rely on a large number of assumptions regarding cellular behavior. One major issue is the degree to which close-to-equilibrium assumptions (such as the dynamics being driven by free energy minimization) can be taken as reliable predictors of the obviously active dynamics of biological cells. The authors have recognized this conceptual issue and have argued that these assumptions provide a reasonable first step for understanding the full complexity of dynamics in the tumor microenvironment.

The problem of T cell infiltration as well as the patterning of the extracellular matrix (ECM) by fibroblasts necessarily involve understanding cell proliferation, cell motion and cell interactions due e.g. to cell signaling. There is evidence that inherently non-equilibrium interactions between the fibroblasts and the extracellular matrix can lead to patterning of the fiber network and trapping of potentially infiltrating T-cells. It is not clear the extent to which this type of interaction can be captured by the approach being used here, although the authors propose that they can be mimicked by proper terms in their formulation. This to me is the primary concern that I had with this paper.

The authors have now addressed what used to be a separate weakness concerning the assumption that fibroblasts affect T cell behavior primarily by just making a more dense ECM. Instead, the organization of the ECM (for example, its anisotropy) could be playing a much more essential role than is given credit for here. This possibility is now discussed in some detail and the authors have suggested that the introduction of a nematic order parameter field would be a useful way to treat this effect.

<https://doi.org/10.7554/eLife.101885.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This is a useful report of a spatially-extended model to study the complex interactions between immune cells, fibroblasts, and cancer cells, providing insights into how fibroblast activation can influence tumor progression. The model opens up new possibilities for studying fibroblast-driven effects in diverse settings, which is crucial for understanding potential tumor microenvironment manipulations that could enhance immunotherapy efficacy. While the results presented are solid and follow logically from the model's assumptions, some of these assumptions may require further validation, as they appear to oversimplify certain aspects in light of complex experimental findings, system geometry, and general principles of active matter research.

We thank the editor for recognizing the usefulness of our work. This work does not aim to precisely describe the complexity of the tumor microenvironment in lung cancer, but rather to classify and rigorously calibrate a minimum number of parameters to the clinical data we collect and generate, and reproduce the global structures of the microenvironment. We identify different scenarios, and show how they depend on the local interactions within this framework. Although we started in the first version with coalescence in the main text and anisotropic geometry in the supporting information, we realized that we needed to provide more directions to better show how our model can be extended. Thus, in Section III-4 we added an analysis of a microenvironment with blood vessels, and showed how to introduce anisotropic friction as a function of fiber orientation, as well as active stress, paving the way for further studies, that would make our model more complex. However, in a first step, it is crucial to start with a limited number of parameters that can be rigorously determined, and this is how this first work was conceived.

Public Reviews:**Reviewer #1 (Public review):**

The authors present an important work where they model some of the complex interactions between immune cells, fibroblasts and cancer cells. The model takes into account the increased ECM production of cancer-associated fibroblasts. These fibres trap the cancer but also protect it from immune system cells. In this way, these fibroblasts' actions both promote and hinder cancer growth. By exploring different scenarios, the authors can model different cancer fates depending on the parameters regulating cancer cells, immune system cells and fibroblasts. In this way, the model explores non-trivial scenarios. An important weakness of this study is that, though it is inspired by NSCLC tumors, it is restricted to modelling circular tumor lesions and does not explore the formation of ramified tumors, as in NSCLC. In this way, is only a general model and it is not clear how it can be adapted to simulate more realistic tumor morphologies.

We thank the reviewer for highlighting the importance of our work. We acknowledge that although we provided anisotropic geometries and the study of the coalescence in the first version, more effort was needed to provide tools to extend our formalism to non-ideal cases. This is now added as Section III-4, where we analyze the impact of blood vessels, and the anisotropic friction due to the nematic order for the fibers; this nematic order can also be used to introduce active nematic stress.

Reviewer #2 (Public review):

Summary:

The authors develop a computational model (and a simplified version thereof) to treat an extremely important issue regarding tumor growth. Specifically, it has been argued that fibroblasts have the ability to support tumor growth by creating physical conditions in the tumor microenvironment that prevent the relevant immune cells from entering into contact with, and ultimately killing, the cancer cells. This inhibition is referred to as immune exclusion. The computational approach follows standard procedures in the formulation of models for mixtures of different material species, adapted to the problem at hand by making a variety of assumptions as to the activity of different types of fibroblasts, namely "normal" versus "cancer-associated". The model itself is relatively complex, but the authors do a convincing job of analyzing possible behaviors and attempting to relate these to experimental observations.

Strengths:

As mentioned, the authors do an excellent job of analyzing the behavior of their model both in its full form (which includes spatial variation of the concentrations of the different cellular species) and in its simplified mean field form. The model itself is formulated based on established physical principles, although the extent to which some of these principles apply to active biological systems is not clear (see Weaknesses). The results of the model do offer some significant insights into the critical factors which determine how fibroblasts might affect tumor growth; these insights could lead to new experimental ways of unraveling these complex sets of issues and enhancing immunotherapy.

We thank the referee for this summary and for recognizing the strengths of our paper.

Weaknesses:

Models of the form being studied here rely on a large number of assumptions regarding cellular behavior. Some of these seemed questionable, based on what we have learned about active systems. The problem of T cell infiltration as well as the patterning of the extracellular matrix (ECM) by fibroblasts necessarily involve understanding cell motion and cell interactions due e.g. to cell signaling. Adopting an approach based purely on physical systems driven by free energies alone does not consider the special role that active processes can play, both in motility itself and in the type of self-organization that can occur due to these cell-cell interactions. This to me is the primary weakness of this paper.

We thank the referee for this important comment, that allows us to clarify this important point. Although biological materials are out of equilibrium, their behavior often resembles that dictated by thermodynamics. Hence the usefulness of constructing a free energy, in terms of these variables. In a first approach to decipher the complex interactions and describe the different and sometimes non-trivial outcomes in this system that involves many components, we must start by minimizing the number of parameters, and identifying those complex processes, that control the evolution of the system. The free energy that we build on this biological system contains therefore out-of-equilibrium processes that can be approximated by a "close to equilibrium" description. Our approach is a classical one in statistical physics of active systems, namely in the effort to construct an equivalent free-energy for out-of-equilibrium systems. This allows to gain a clearer insight into those complex processes.

We have added a sentence in the main text, section III.1, to clarify this point:

“Building a free-energy density for a biological material is justified, because, although biological materials are out of equilibrium, their behavior often resembles that dictated by thermodynamics. It is therefore useful to write a free energy in terms of state variables.”

Nevertheless, we recognize that we should have provided more tools for using our formalism by making it active. This is why we introduced the nematic order in the fibers in Section III-4. This nematic order can be used to introduce active stress, and we have cited previous works by some of us see [?, ?, ?] as references for building active processes out of it.

We must also note that cell signaling has been introduced *a minima* in our system for providing the cue for the arrival of T-cells and NAFs from the boundaries. However, we found that although we had evoked the other role of the chemicals in the transformation from NAFs to CAFs in the text, details were not well explained. We have therefore corrected and added some explanations in the introduction of section III, and III.1, III.2.

A separate weakness concerns the assumption that fibroblasts affect T cell behavior primarily by just making a more dense ECM. There are a number of papers in the cancer literature (see, for some examples, Carstens, J., Correa de Sampaio, P., Yang, D. et al. Spatial computation of intratumoral T cells correlates with survival of patients with pancreatic cancer. Nat Commun 8, 15095 (2017); Sun, Xiuji, Bogang Wu, Huai-Chin Chiang, Hui Deng, Xiaowen Zhang, Wei Xiong, Junquan Liu et al. " Tumour DDR1 promotes collagen fibre alignment to instigate immune exclusion." Nature 599, no. 7886 (2021): 673-678) that seem to indicate that density alone is not a sufficient indicator of T cell behavior. Instead, the organization of the ECM (for example, its anisotropy) could be playing a much more essential role than is given credit for here. This possibility is hinted at in the Discussion section but deserves much more emphasis.

The referee is right in his comment, and we thank him for raising this issue. We have therefore introduced the anisotropic orientation of the fibers, which induces an anisotropic friction in a new section III-4. In addition, the references pointed out were included in this section. However, although the anisotropy strongly influences the fate of the tumor when the fibers are oriented perpendicular to the surface of the cancer nest, it is less effective when the fibroblasts are oriented in the direction of surface of the cancer nest. In the latter case, which is often the case before cancer cells reshape the tumor microenvironment, the matrix density should correlate with the friction.

Finally, the mixed version of the model is, from a general perspective, not very different from many other published models treating the ecology of the tumor microenvironment (for a survey, see Arabameri A, Asemani D, Hadjati J (2018), A structural methodology for modeling immune-tumor interactions including pro-and anti-tumor factors for clinical applications. Math Biosci 304:48-61). There are even papers in this literature that specifically investigate effects due to allowing cancer cells to instigate changes in other cells from being tumor-inhibiting to tumor-promoting. This feature occurs not only for fibroblasts but also for example for macrophages which can change their polarization from M1 to M2. There needed to be some more detailed comparison with this existing literature.

The referee is right that the first part of our approach, namely the dynamical system may be common in this kind of system, and it needs to be mentioned. So we added the following sentence in the discussion: "This is in line with several similar mathematical models, that study through this lens the inhibition/activation of the immune system by cancer cells either by means of compartmental nonlinear models similar to our dynamical system, for instance regarding macrophage recruitment and cytokine signaling {arabameri2018structural} {li2019computational}, or mixture models {fotso2024mixture}. We combine the two

approaches in order to rigorously derive the parameters of the model and gain insights from both.”

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

The authors should address the following points:

Major issues

(1) The shape of tumors simulated differs immensely from the observed tumors in Fig. 2. Here, the tumor is constituted by irregular domains, not dissimilar from domains in phase separating mixtures. The domains simulated are circular. Since the authors are using the space dependent model to model the increase in tumor cells with time in the different scenarios (immune-desert, immune-excluded, immune inflamed), it should explain how non-spherical tumor structures can be observed in these scenarios. The authors introduce tumor coalescence in page 28, however, it is not expected that the structures observed in Fig 2 are the result from different tumors merging and coalescing, because that would result from an unlikely large number of initial mutation events in the same region of the tissue. The authors should explain what mechanisms present in the model can lead to non-spherical forms.

We agree with the reviewer that real tumors are rarely round contrary to what our numerics suggests. In fact, only the last figure of our paper in the supporting information was more appropriate for such a discussion. We are now adding discussions and new figures to better illustrate our spatial model, see Figure 6 and section III-4. The in situ geometry of tumors depends on the shape of the host organ, the diffusive (chemical) or advected species such as T cells and fibroblasts, and on the nutrients. Thus, in our case, only cancer cells are produced locally, but during growth the tumor is strongly constrained by the microenvironment, and thus the geometry of the domain we model in the numerics and its boundary conditions. This is also true for the chemicals responsible for growth, cellular advection and phenotypic transformation. Their concentration depends on a convection-diffusion equation and boundary conditions. For a tumor in situ, such as in the lung, the available space is a constraint that will dominate the final geometry of the tumor nests. We do not think that coalescence is controlled by mutational events, but most likely by the search for space necessary for growth. Compared to the first version, we add new figures (Figure 6) that show that the geometry of the organ, as well as the localization of blood vessels, are a cause of the irregularity of the tumor shapes. We also introduce orientational order, which as suggested in section III-4, can induce anisotropic friction and stresses, as well as anisotropic growth. We cite (Ackermann, Joseph, and Martine Ben Amar. "Onsager's variational principle in proliferating biological tissues, in the presence of activity and anisotropy." The European Physical Journal Plus 138.12 (2023): 1103.) where we described active stresses and coupling related to anisotropic growth.

(2) According to the authors, the model presented in equations (1) and onwards simulates the evolution of the fraction of tumor cells in the tissue. However the fraction of tumor cells, for example, depends itself on the variation of other cell types. For example, if fibroblasts were to proliferate with rate α , even without tumor cells proliferating, the fraction of tumor cells in the mixture should decrease as α times the tumor cells fraction. These terms are missing. The equations do not describe the evolution of the cells' fractions but of the amount of cells of each type, normalised by the total carrying capacity of non-normal cells in the tissue. The text should be rewritten accordingly.

We agree with the referee: our definition of cell density was not precise enough and may appear misleading. In the paragraph II1, we more explicitly introduce the word mass fraction which is the correct physical quantity to introduce into the spatial model.

"All these cells have the same mass density and the sum of their mass fraction satisfies the relationship $S = C + T + F_{NA} + F_A = 1 - N$, where N is a healthy non active component as healthy cells, for example."

It is less intuitive than "number of cells per unit volume" but necessary for the following (III)

(3) The authors start by calculating fixed points of different versions of the dynamical system without spatial dependence. They should explain what is the relevance of these fixed points: in a real situation, where the concentration of tumor fibroblasts and T-cells depend on position, in which conditions are these fixed points relevant?

The referee is right and we will clarify this point: the dynamic analysis is a help for understanding and predicting the scenario occurring in the system. After all the steps of paragraph 2.2, we are faced with 11 independent parameters only for the dynamical system and without the parameters generated by the space modeling itself. Our estimation concerns only lung cancer. These parameters do not appear in the literature. The parameters introduced in Sec. III which are more related to physical interactions such as friction, cell-cell adhesion, etc. can be found in the literature or can be estimated and thus measured in in vitro experiments (see Ackermann and Ben Amar, EPJP 2023, P. Benaroch, J. Nikolic et al. 2024, biorxiv). So what are the fixed points for: they help to get the right numbers for spatial analysis. To recover special features of cancer evolution, we need a model, but also correct estimates of the data in a code that is quite technical and heavy, with each simulation taking a certain amount of time. For users who only need rough predictions, the analysis in section 2 is sufficient.

It is also important to note that the global result depends only on the source terms, and on the boundary conditions. This can be illustrated with a simple example: Consider the governing equation for the density of a component with velocity $\mathbf{v}$ and source term:

$$\partial_t \phi + \nabla \cdot (\phi \mathbf{v}) = \gamma \quad (1)$$

Integrating the equation over a fixed volume V of surface S gives:

$$\partial_t \Phi = \Gamma - J_S = \bar{\Gamma} \quad (2)$$

$\Phi = \int \phi dV$, $\Gamma = \int \gamma d\Omega$, $J_S = \int (\rho \mathbf{v}) \cdot \mathbf{n} dS$. This integrated equation can then be

approximated by the dynamical system that we write. Thus, while the dynamical system does not give any information about the local structure of the system, it may be indicative of its global outcome.

(4) In page 15, the authors identify that α_{NA} is proportional to $\delta \epsilon^4$. However, in equation (7), they replace α_{NA} by $\delta \epsilon^4$ without the proportionality constant. This should be corrected.

Thank you for your remark. This typo is now corrected.

(5) The tumor cell movement should be much slower than the T-cells. Here, the authors assign a similar friction coefficient for the cancer cells and T-cells, for example. However, in lung cancer tumor cells are epithelial, and adhere to each other in the tissue. Their movement is very restricted by the basement membranes and by cell-cell adhesion. Immune cells and T-cells on the other hand move rapidly throughout the stroma. It is a gross simplification to not consider the low epithelial tissue mobility in the context of lung cancer.

It is possible to assume different friction coefficients for each phase pair. This has been done in a previous publication, Ackermann et al., Physics report 2021. It is also possible to play with the cell-cell adhesion in the energy density and on the diffusion coefficient introduced in the Flory-Huggins free energy. Cell-cell adhesion is taken into account in the energy, and this makes the tumor a more dense phase, while T-cells can move towards cancer cells to which they are attracted. In the last part of the paper, we show the role of an anisotropic friction due to a nematic order for activated fibroblasts and all the other cells

(6) What is the biological mechanism by which the T-cells form a colony with a surface tension? In the phase-field model, the authors have a surface tension assigned to the cancer cells, T-cells and fibroblasts. Can the authors justify biologically why do they consider these surface tensions?

The fact that T-cells form a colony is due to the accumulation of T-cells at the outer boundary of the tumor, as they are attracted to it but cannot penetrate due to the strong cell-cell adhesion of the tumor cells in the nest. Adding a gradient square is standard in continuous models to limit the sharp variations. In a continuous approach, the gradient square contribution limits the sharp variations in cell density which are not physical.

Minor issues

(a) Page 6 (end), characterisation of the fibre barrier produced by CAFs missing: what is the fibre density, how it can hinder the spread of cancer and T-cell motility? Is it so dense that it prevents amoeboid movement? Can cells move through it using matrix degradation proteins?

The fiber density corresponds to the fibrous organic extracellular matrix secreted by cancer-associated fibroblasts. In desmotic (highly fibrous tumors such as PDAC or NSCLC), this extracellular matrix deposited around the tumor forms a physical barrier around the tumor nest, preventing both cell migration and capillary and immune cells penetration. In these cases, the fibrous belt actually prevents amoeboid movement and cells must deform significantly to migrate. The role of this barrier was particularly demonstrated in the reference (Grout, John A., et al. "Spatial positioning and matrix programs of cancer-associated fibroblasts promote T-cell exclusion in human lung tumors." *Cancer Discovery* 12.11 (2022): 2606-2625.). In later stages of cancer, the tumor may adapt and develop strategies to metastasize, such as matrix degradation. This matrix can be oriented, organized or disordered. To build a minimal model, we first considered an isotropic friction and also an anisotropic friction of the nematic belt, due to the activated fibroblasts. In the case of T-cells, as mentioned in section I.1, it is true that the biological literature also considers a phenotypic transformation of the T cells by the activated fibroblasts: this concerns both their proliferative capacities, antigen recognition and also their cytotoxic function. To better document the different mechanisms, we add the following publication: Cancer associated fibroblasts-an impediment to effective anti-cancer T cell immunity, by Koppensteiner, Lilian and Mathieson, Layla and O'Connor, Richard A and Akram, Ahsan R, *Frontiers in immunology* (2022).

However, our goal is to build a minimal model and to characterize and quantify the physical process in which CAFs are involved, namely the role of a physical barrier, that has been documented, as documented above.

(b) Page 19 (Fig 3), in the figure legend it is written "resting fibroblasts", should be "non-activated fibroblasts".

The referee is right: it will be better to write non-activated fibroblasts. This is now changed in the main text.

(c) Page 21 (equation), what is $d\Omega$? It is dr ?

We thank the referee for raising this point. The text was indeed ambiguous as sometimes $d\Omega$ was replaced by dr . To be clearer, all the elements of volume are now noted dV , and the element of surface of the system are noted dS .

In the article the units are in italic and should be in roman.

Thank you for raising this point. It has been corrected.

(d) Page 25 (beginning section III.3), the authors mention that the simulation is 2D, however, the simulation has radial symmetry. A 1D simulation in radial coordinates could simulate a 3D spherical system. Is the simulation of this section equivalent to a 1D radial simulation (in 2D)?

The referee is right that in radial symmetry, a 1d equation may be written. We therefore present numerics with irregular shapes of the tumor nest in order to make the system fully 2d.

(e) Page 26 (Fig 4). Legends inside the plots of plates A, B, C and D are not clear. Colorbar range of plates A and D is different. Would facilitate if the ranges were the same.

The referee is right: the surface plots presented in figure 4 would be easier to compare with the same colorbar range for the legends. In fact, as the referee noted, figures in A, B and C have the same legends, while figure in D has a different one. This is due to the fact that D represents the case of the immune-inflamed tumor where the cancer mass fraction is quite vanishing, resulting in values that are of 3 orders of magnitude lower than those present in A, B and C. Therefore, they would disappear if the colorbar range were equal to the others. We insist more on the change of scale in the legend of Figure 4, in the new version.

(f) Page 29 (Fig 5), would facilitate if the order of immune-desert, immune-excluded, immune-inflamed was maintained throughout the document. In this figure the immune-inflamed case appears first.

We agree with the reviewer that following the same order in which the different cases are presented throughout the manuscript would be helpful in comparing the different figures. Therefore, we have modified Figure 5.

(g) Page 31, the authors indicate that pharmacodynamics and pharmacokinetics are highly dependent on tumour spatial structure. Can they provide examples and citations?

In the discussion, we have added references concerning pharmacodynamics.

(h) Page 33 (Fig Sup2), would facilitate if the order of immune-desert, immune-excluded, immune-inflamed was maintained throughout the document. ±±

We thank the reviewer for pointing this out, the order of the different scenarios in Fig Sup 2 has now been changed.

Reviewer #2 (Recommendations for the authors):

Major points

(1) Following on from the discussion in the public review, I feel that there are a number of critical issues that need to be addressed regarding modeling assumptions. I would like to understand why the authors believe it is possible to use a free energy-driven model of the microenvironment when many of the processes relevant for their study have an undeniably "active media" flavor.

The referee is right that processes in biology are active processes. However, it is a classical approach to model physical interactions between biological components with a free-energy, especially cell adhesion, as they often lead to quasi-stationary equilibrium-like patterns. The free-energy approach has also the advantage to derive straight-forwardly complex phenomena involving many components. Activity can indeed be introduced in such a framework, if we know that the fibroblasts transform into myo-fibroblasts, see for example our previous publication Ackermann and Ben Amar, EPJP 2023. However, in the interest of simplification and reduction of the number of free parameters, we have not not considered further complication of the model here, as a minimal model allows to distinguish the main processes that occur. Nevertheless, introducing more precisely activity, in the nematic approach already achieved for the friction, is a natural continuation of our work: See the new Section III-4, where we introduce the nematic order, and we indicate that active nematic stresses can be written from it.

Next, I don't understand the assumption that T cells do not proliferate once they detect neoantigens on the cancer cells; activation of T cells usually causes them to become more proliferative.

We thank the referee for this question. The T-cell fraction has two origins: proliferation of T-cells in situ in the stroma or inside tumor nest or external arrival from the sources that we privilege. We recognize that a full analysis of the tumor-microenvironment would require to consider proliferation near the tumor, as many more other processes which is do able but requires the knowledge of more biological date. In addition, besides, the proliferation of T-cells will be equivalent to increase the killing abilities of T-cells and these two effect overlapp in our approach.

In order to clarify this point, we modify the following sentence in Section II.2:

"Although proliferation of cytotoxic T-cells has been observed, we do not consider explicitly proliferation in our study as we focus on their ability to infiltrate the tumor."

Rather, we consider that T-cells proliferate outside the domain boundaries, so that this proliferation is included in the boundary source contributions.

Finally, the issue of whether the density of fibers is sufficient to understand the role of fibroblasts is not at all settled. There should be a full discussion of this issue including mentioning of the Nature paper (cited in the public review) that argues that orientation (and not density) is the key to the role of fibers, as well as the earlier cited work of Kalluri and collaborators on the role of ECM density in pancreatic cancer.

We thank the referee for this remark. As we wrote above in the response to the public review, we introduced significant additions that aim to tackle this question in the article.

(2) The authors present a picture of a tumor cell with fibroblasts apparently arrayed circumferentially around the tumor boundary and therefore blocking infiltration. This type of tumor structure has been seen before, for example in "On the mechanism of long-range orientational order of fibroblasts." Proceedings of the National Academy of Sciences 114, no. 34 (2017): 8974-8979, which should be cited. More importantly, in that paper the argument is made that positive feedback between fibroblasts and ECM geometry can cause structures like this to form. If this is indeed what is occurring, this would indicate the crucial importance of a mechanism beyond what is contained in the current model. This issue should therefore be discussed within this paper. This issue is of course connected to the previous point regarding the role of ECM structure beyond density.

We completely agree that the interplay between the fibroblast layer and the tumor shapes the tumor boundary. One of the authors has worked recently on this precise topic (Aging and freezing of active nematic dynamics of cancer-associated fibroblasts by fibronectin matrix remodeling, C Jacques, J Ackermann, S Bell, C Hallopeau, CP Gonzalez, ... bioRxiv, 2023.11.22.568216, Ordering, spontaneous flows and aging in active fluids depositing tracks S Bell, J Ackermann, A Maitra, R Voituriez arXiv preprint arXiv:2409.05195). Since the fibroblast layer is an active material, it contributes to an anisotropic stress that can be introduced into the model. Our first strategy was to present the simplest modeling in order to focus on the most important interactions as cell-cell adhesion and cell-tissue adhesion. However, we recognize that those questions should be discussed in the text, and we discuss it in the new section III-4

Minor points

There are also a number of more minor points to consider:

(1) Since the parameter is taken to be $O(1)$, why exactly does it matter how the other parameters scale with it?

It is very important to compare the order of magnitude of the other parameters once the selected parameter of order $O(1)$ is really the driving parameter of the coupling. It gives a first picture of the main interactions that has to consider.

(2) I didn't understand the relevance of referring specifically to IL 6 among many other possibly relevant signals, as is currently done on page 7.

This corresponds to studies aiming to correlate lung cancer risks and the concentration of interleukin, mostly IL6 and IL8 (McKeown, D. J., et al. "The relationship between circulating concentrations of C-reactive protein, inflammatory cytokines and cytokine receptors in patients with non-small-cell lung cancer." British journal of cancer 91.12 (2004): 1993-1995., Brenner, Darren R., et al. "Inflammatory cytokines and lung cancer risk in 3 prospective studies." American journal of epidemiology 185.2 (2017): 86-95.) but in the absence of very detailed biological information, the modeling and its results are not modified if other chemicals intervene..We slightly modified the following phrase in section I.1:

"In particular, in the family of inflammatory proteins, also called cytokines, Interlukin-6 (IL6) and (IL8) seem, among others to stimulate the infiltration of $CD8^+$.

(3) The authors need to mention the possibility of T-cell chemotaxis to the tumor being "self-amplified" in the T cell system, as put forth in Galeano Ninõ, Jorge Luis, Sophie V.

Pagoon, Szun S. Tay, Feyza Colakoglu, Daryan Kempe, Jack Hywood, Jessica K. Mazalo et al. "Cytotoxic T cells swarm by homotypic chemokine signalling." *eLife* 9 (2020): e56554. This might again reveal a needed extension of the current modelling strategy.

We thank the referee for his/her comment on the self-amplification of T-cell population in the stroma and we mention the indicated reference in our paper. This auto-chemoattractant process which induces a dynamic of more efficient recruitment towards the tumor, may be important for immunotherapy. To have more efficient T-cell arriving at the site of the tumor, will lead a better issue for the patient, if the swarming organization is maintained in a desmoplastic nematic stroma.

(4) It is not obvious to me that in sub figures 3F and 3H the tumor is enroute to being totally eradicated, as is stated in the text. The blue lines seemed to asymptote at non-zero population values.

Looking at sub-figures 3F and 3H, we stated in the main text that the tumor is eradicated as the representative population approaches a 0 value fraction, or at least decays around the 0 (0.01/0.05 to be more precise). This is even more evident when compared with the other cases where the tumor mass fraction reaches values of a higher order (up to 0.6), thus leading us to distinguish between these different scenarios.

(5) The description of the interaction of cells with fibers as being increased friction might be misleading, as the real effect could be actual trapping in the network (as opposed to just slowing down the motion).

We thank the referee for this question as it allow us to make an important distinction. Indeed, what the referee describes seems to correspond to a discrete event, namely a cell trapped in a network. However, coarse-graining the dynamics to the continuous modeling seems to us as leading to an effective friction between the two phases. Moreover, we also now introduced an anisotropic friction which can represent a trapping. The velocities are not only directed around the tumor but can also be oriented towards the tumor, so that eventually the friction along the radius mimics a trapping (see Fig.4 on top). We have introduced this anisotropic friction via a nematic model, see the appendix.

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